

AI in Power Plants

From Data to Decisions

An introduction to Artificial Intelligence for smarter plant operation and maintenance

PREDICT • DIAGNOSE • OPTIMISE • ASSIST

MAHAGENCO Training Centre, Nashik

Full-day programme · 10:00–13:00 and 14:30–17:00 · in person at Nashik, with participants online from Koradi, Khaperkheda, Bhusawal and Paras

PRESENTER

S. H. Parihar

Power sector engineer · author · AI practitioner

AUTHOR OF

- *Lean AI*
- *Learn the English that AI Understands*
- *AI for Busy Parents*

16 years

experience in the power sector

What you will take away

Session objective — one working day, four things you keep

1

A working mental model of AI

What machine learning, generative AI and an agent actually are, described in plant language rather than in vendor language.

2

Where it earns its keep in a power station

Availability, heat rate, auxiliary power, coal quality and knowledge — the places where a thermal station has real money on the table.

3

An honest view of the limits

What AI gets wrong, where it must never be connected, and why most programmes fail for reasons that have nothing to do with the algorithm.

4

A first step you can take at your own station

One pilot, chosen from your own June 2026 numbers, that a named engineer can run in ninety days without a fleet-wide sanction.

The day, hour by hour

Nine sessions, two tea breaks and lunch — with the simulation lab running through the middle of it

1	AI fundamentals	10:10–11:10
•	Tea	11:10–11:25
2	Intelligent monitoring	11:25–12:10
3	Predictive maintenance	12:10–13:00
•	Lunch	13:00–14:30
4	Reading our own numbers	14:30–15:00
5	What other generators have done	15:00–15:12
6	Performance optimisation	15:12–15:45
•	Tea	15:45–15:55
7	Generative AI, RAG and agents	15:55–16:28
8	Computer vision and emerging	16:28–16:40
9	Implementing responsibly	16:40–17:00

8 hands-on

simulation modules that run offline, in a browser, on the laptop in front of you

Breaks

Morning tea	11:10 – 11:25
Lunch	13:00 – 14:30
Afternoon tea	15:45 – 15:55

How today works — the simulation lab

Eight modules you run yourself, and one boundary that applies to all of them

● One file, no internet

A single HTML file that opens in any browser on your own laptop. No installation, no login, no network. It works with the cable unplugged.

● Real algorithms, stated data

Nearest-neighbour residuals, ridge regression, TF-IDF retrieval — genuine code, not animation. Data is either our actual June 2026 figures or synthetic data whose physics equations are printed on screen.

● You keep the file

Copy it to a pen drive and take it away. Re-run every module at your own station, with your own people, whenever you want.

● A teaching instrument, not a plant model

These modules are not calibrated to your unit. Nothing that appears on these screens today should ever influence an operating decision.

Real algorithms, honest data, no plant authority. Learn the method here, and validate every result against your own engineering before it means anything.

1

SESSION

AI fundamentals for power plant engineers

What AI, machine learning and generative AI actually are — in plant language, not vendor language

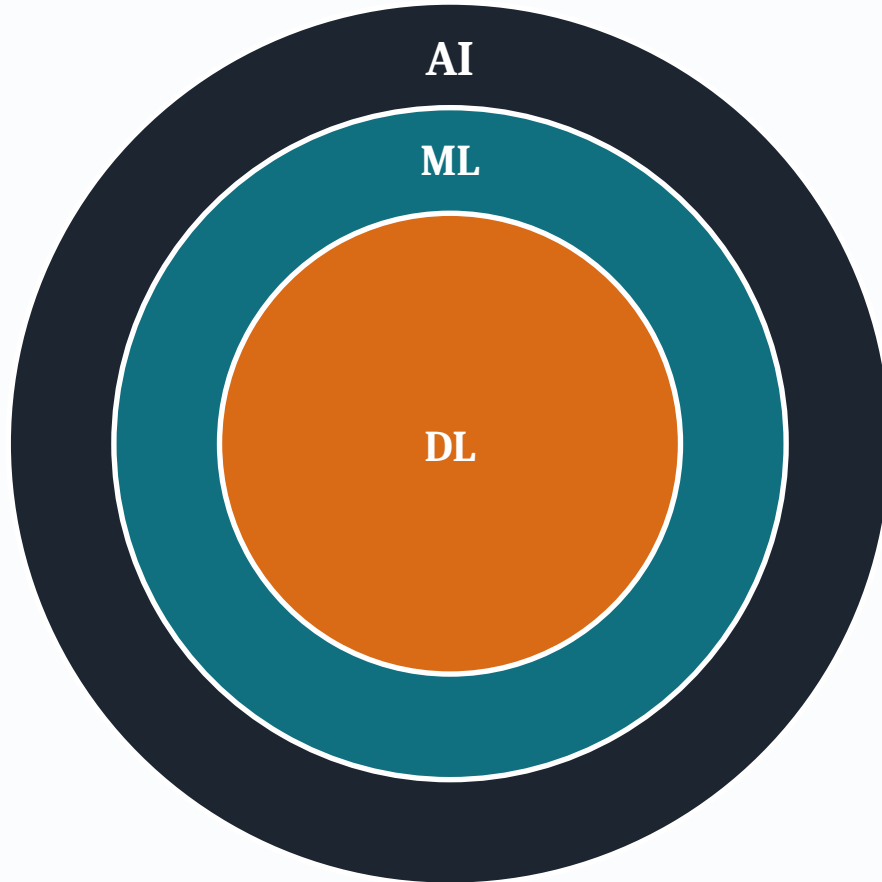
- The one idea that matters: expected versus actual
- How a machine learns from plant data, and how it is used
- Inside a neural network, without the mathematics
- How to judge a model the way an operator will
- What AI is not — and the data pathologies that break it

10:10 – 11:10

60 minutes

What AI actually is

Four words vendors use interchangeably, and why the difference matters to you



Artificial Intelligence

Any technique that makes a machine do something we would call intelligent. Rule-based expert systems count. The interlock logic in your DCS is, strictly, the oldest form of it.

Machine Learning

The machine works the rule out from data instead of being handed it. Almost every application in today's course sits inside this ring.

Deep Learning

Machine learning using large neural networks. Needed for images, sound and language. Usually overkill for process signals out of a historian.

Generative AI

Beside the circles, not inside them. Models that produce new text, images or code. Excellent with manuals, SOPs and reports. Knows nothing about your unit until you give it your documents.

The asset you already own

What a 210 to 660 MW unit holds today, with no new investment at all

Source	What it holds	Typical volume (indicative)	What it is used for today
DCS and historian	Pressures, temperatures, flows, MW, damper and valve positions	Thousands of tags per unit	Live screens and trends
Vibration and condition monitoring	Shaft and bearing vibration, orbits, spectra	Continuous on TG, route-based elsewhere	Alarm and periodic review
CMMS / SAP PM	Work orders, defect history, spares, notifications	Years of records per station	Compliance reporting
Chemistry lab / LIMS	Cycle chemistry, coal and ash analysis	Daily to hourly	Statutory records
CEMS	Stack emissions and opacity	Continuous	Regulatory submission
Coal and fuel accounting	Rake weights, loading and unloading GCV, transit loss	Every rake, every day	Bill preparation
Documents	Manuals, SOPs, RCA and trip reports, drawings	Decades of paper and PDF	Found when someone remembers it exists

4,588 MU

sent out in June 2026 — recorded second by second, and almost none of it ever re-read

The one idea that matters

Expected versus actual — the residual

THE QUESTION A FIXED ALARM ASKS

Is it above 90 °C?

One tag, one number, no context. The same 90 °C setpoint at 55 per cent load on a January night and at full load on a May afternoon. It cannot tell the difference, and it was never asked to.

THE QUESTION A MODEL ASKS

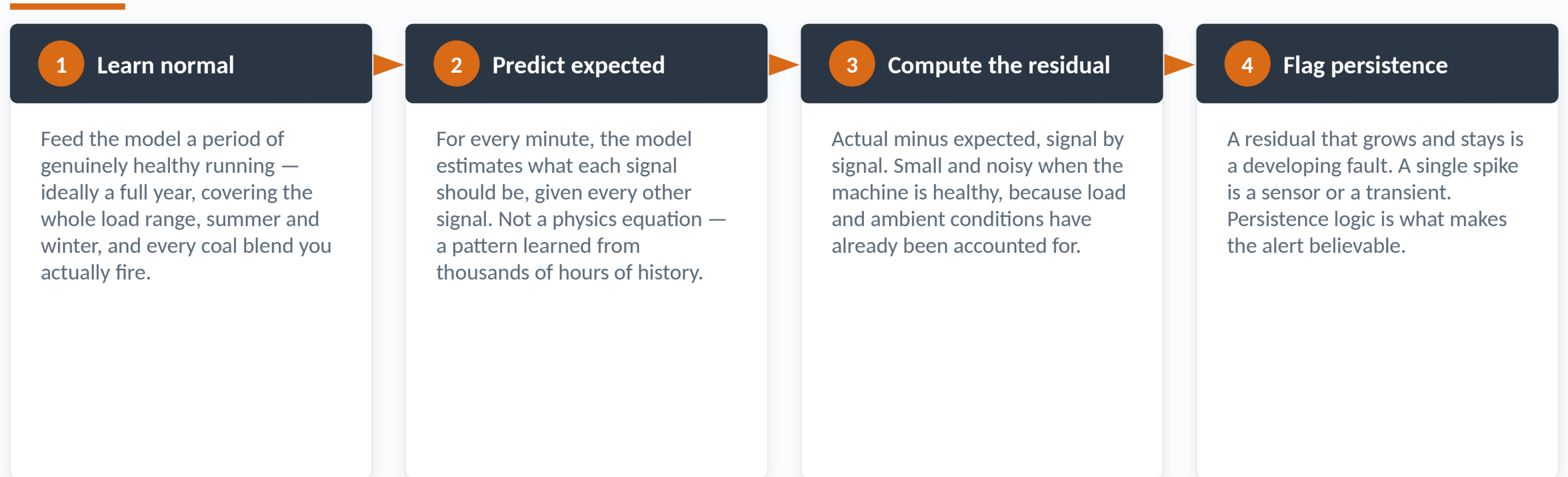
Is it above what it should be?

Given this load, this ambient, this cooling water inlet and this oil flow, the model says what the bearing should read right now. Subtract that from what it does read. What is left over is the only part that carries information about the machine.

Actual minus expected is the residual. Every process-data application in this course is built on that one subtraction — and Session 2 walks a real bearing through it week by week.

How a machine learns from plant data

The same four steps sit behind almost every application you will see today

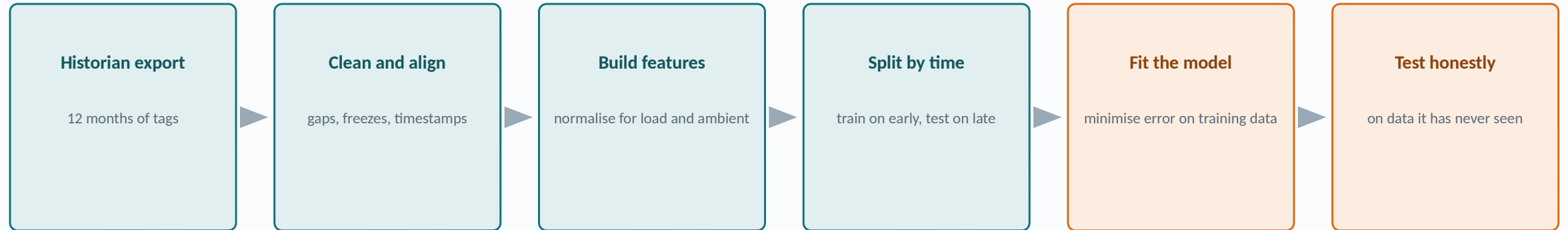


Step one decides everything. Train on a period that already contained the fault and the model learns the fault as normal.

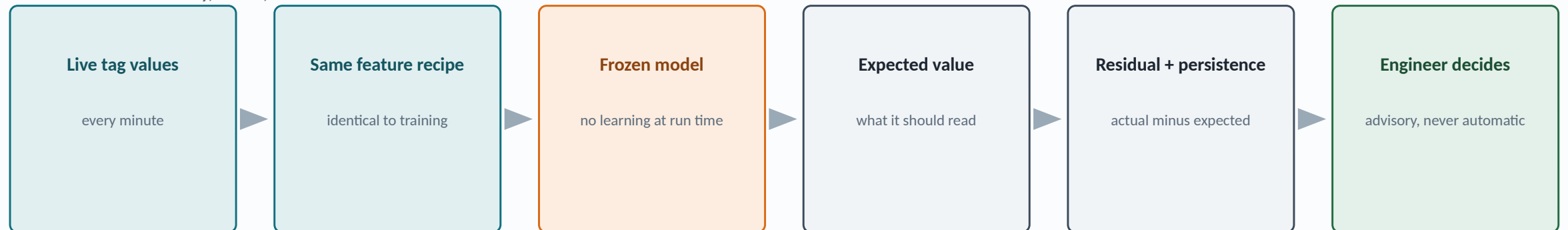
How a model is built, and then used

Two separate pipelines that people constantly confuse — training happens once, inference happens every minute

TRAINING — once, offline, with an engineer curating



INFERENCE — continuously, online, unattended



Feedback: Retrain when the machine changes: after an overhaul, a retrofit, or a change of coal source

Three ways machines learn

Two plant examples each — and where the value actually sits for us

Family	What it needs	Plant example 1	Plant example 2
Supervised — learns from labelled examples	History where you already know the answer	As-fired GCV predicted from plant signals, with lab results as the answer key	Vibration signature classified as unbalance, misalignment or looseness
Unsupervised — finds structure with no labels	Only a clean period of healthy running	ID fan or mill anomaly detection, with no failure history at all	Grouping operating states to spot an unusual mill combination
Reinforcement — learns by trial and consequence	A simulator, or a very tolerant process	Combustion setpoint search, in a simulator only	Not appropriate for direct control of a utility boiler today

Most plant value today comes from the middle row. It needs no failure history — only one clean period of healthy running.

A tag is not a feature

Why raw historian values go into a model badly, and what engineering fixes them

Raw tag, as the historian stores it

- Bearing temperature, 74 °C — meaningless without load and ambient
- Mill current, 62 A — depends entirely on coal flow and fineness
- Condenser back pressure — moves with CW inlet temperature all year
- APH outlet gas temperature — moves with load, coal and air flow
- One instantaneous value, with no memory of the last six hours

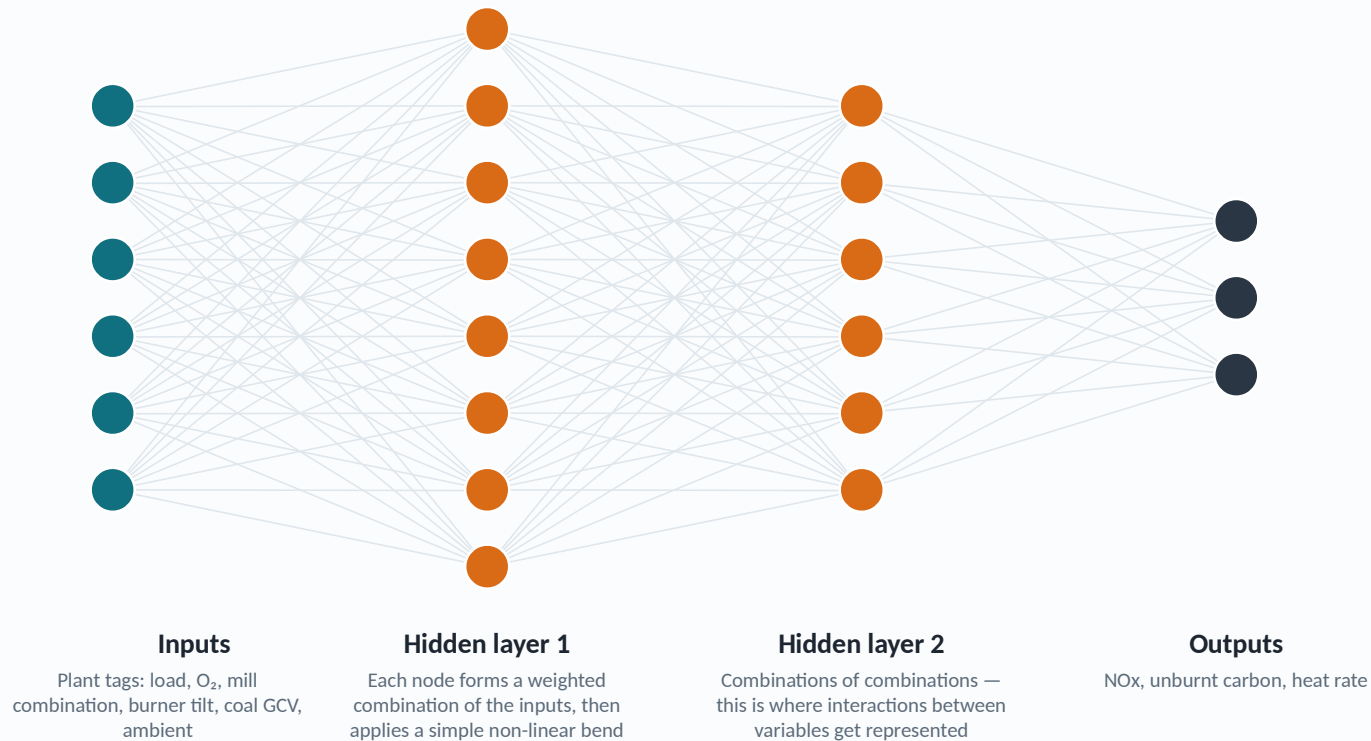
The feature an engineer builds from it

- Temperature rise above cooling medium, normalised for load and ambient
- Current per tonne of coal — a ratio that holds across the load range
- Back pressure minus achievable at today's CW inlet — the controllable part
- Gas-side and air-side differential, and the leakage ratio derived from it
- Rolling mean, standard deviation and slope over one, six and twenty-four hours

Feature engineering is where plant knowledge enters the model. It is the one part a data scientist cannot do without you.

Inside a neural network, without the mathematics

What is actually happening between the inputs you know and the output you want



What is actually happening

- A 'weight' is just a multiplier the machine adjusts. Training means nudging every weight, thousands of times, until the predicted outputs match history.
- The non-linear bend at each node is what lets the network represent a curve rather than a straight line. Without it the whole network collapses to simple linear regression.
- Nothing inside is a physical quantity. The hidden nodes do not correspond to anything you can point at in the boiler — which is exactly why explainability tools matter.
- For most plant process data a network this small is enough. Deep networks with millions of weights are for images, sound and language, not for forty tags.

A neural network is a very flexible curve fitted to your history. Its power and its danger are the same thing: it will fit anything you give it, including nonsense.

Training, validation, and the cardinal sin

How you know a model works before you ask an operator to trust it

● Training data

The period the model learns from. Known-good running, wide enough to cover every load, season and fuel the unit actually sees.

● Validation data

A period the model has never seen. If it performs here it has learned the machine. If not, it has memorised the training set.

● The cardinal sin

Judging a model on data it was trained on. It will look magnificent and fail in week one. Almost every inflated vendor claim starts here.

● Time-based split, never random

Plant data is correlated minute to minute. A random split leaks the future into the past. Train January to September, test October onwards.

Ask any vendor one question: what period was the model tested on, and had it seen that period before?

Overfitting, explained physically

The difference between a model that learned the machine and one that learned last year

A MODEL THAT MEMORISED

It learned last year's noise

It reproduces the training period almost perfectly, including the transmitter drift, the odd monsoon fortnight and the month one mill was out. Show it this year and it flags normal running as abnormal, while missing the real drift entirely.

A MODEL THAT GENERALISED

It learned how the machine behaves

Slightly worse on the training period, and far better on data it has never seen. It survives a coal change, an overhaul and a season it was not trained on, because it learned the relationship rather than the record.

A model that fits history perfectly has learned history, not the machine. Suspicion, not admiration, is the correct response.

Judging a model the way an operator will

The metric on the vendor's slide is not the metric that decides whether it survives

What the data scientist reports

R^2 of 0.98 on the test set

Root mean square error in degrees

Model accuracy of 94 per cent

Area under the curve

Trained on two years of data

A very good model nobody acts on

What the shift engineer actually asks

How many weeks of warning did it give?

How many false alarms will I chase in a month?

When it fires, is there something to go and look at?

Does it tell me which signal moved, and why?

Is it still right after the overhaul?

A merely adequate model that changed a decision

Lead time, false alarms per model-month, actionability. Those three decide adoption. R^2 decides nothing at all.

What confidence means, and what it does not

A number without an error bar is an opinion with a decimal point

● Confidence is a range, not a promise

A useful output says the bearing should be 68 °C plus or minus 1.5 °C. The range carries as much information as the number.

● Two very different uncertainties

The model is unsure because the data is noisy, or because it has never seen this operating point before. The second is far more dangerous.

● Calibration

If a model says ninety per cent confident on a hundred occasions, roughly ninety should turn out right. Untested, that percentage means nothing.

● Outside the training envelope

At a load, ambient or coal the model never saw, it will still produce an answer. It will be confident, and it will be wrong.

● Confidence is not correctness

A generative model will state an invented clearance value with complete assurance. Assurance and accuracy are entirely unrelated quantities.

● What to demand

Ask for the error band, the coverage of the training data, and the operating range where the vendor will not stand behind the output.

Any output arriving without an uncertainty attached is an opinion, however many decimal places it happens to carry.

Explainability — an alert with its reasons attached

No engineer acts on a number with no justification, and they are right not to

Koradi Unit 9 · BFP 6B · Advisory · 06:40

● Signal	Bearing 2 temperature, drive end
● Deviation	68.4 °C measured against 64.9 °C expected — plus 3.5 °C
● Persistence	Above threshold continuously for 9 days
● Largest contributor	Thrust bearing temperature rise per unit of fan power — 61 % of the deviation
● Second contributor	Lube oil outlet temperature, rising alongside it — 22 %
● Recommended action	Check oil condition and cooler performance at next opportunity — not an emergency

The contribution breakdown is what converts an alert into a work order. Without it you are asking an engineer to trust a number.

Where AI genuinely helps in a plant, ranked

An honest view — buy the top, pilot the middle, only read about the bottom

Anomaly detection on rotating plant	Model of healthy behaviour, residual on every signal, persistence logic before any alert is raised	Proven — the safest place to start
Soft sensing	Predicts a value you cannot measure online from values you can, trained on lab results	Proven — needs honest error reporting
Diagnostics	Points at which signals drove the deviation and matches the pattern to known signatures	Emerging — good at pointing, weak at naming
Prognostics and remaining life	Fits a degradation curve to the residual and projects a date with an uncertainty band	Emerging — the band is usually wide
Advisory optimisation	Searches setpoint and equipment combinations, recommends, and leaves the operator to decide	Proven as advisory — gain depends on current tuning
Closed-loop control	Writes bounded bias values to the DCS under clamps, rate limits and a watchdog	Emerging — only with full engineering controls
Vision and language	Cameras reading leaks, gauges and safety; retrieval over manuals, SOPs and trip reports	Proven for document search · Emerging for vision

Start where the answer is a number an engineer can check. Anomaly detection and soft sensing repay their cost before anything else does.

What AI is not

Say all five of these out loud before anyone in this company buys anything

Never in protections

Advisory and statistical, sitting above your interlocks — never inside them. No model output belongs in a trip circuit or a safety function. Not now, not in five years.

Not a judgement engine

It produces a number and a reason. Weighing risk, reading the plant, and deciding what to do on a running unit remain entirely the engineer's work.

Not magic without data

No historised tag, no model. A frozen transmitter, an over-compressed trend or a tag that was never recorded cannot be rescued by any algorithm.

Not a one-time project

A model is a maintained asset. After an overhaul, a retrofit or a sustained coal change it must be retrained, or it will quietly become wrong.

Not a headcount saving

It covers equipment nobody has time to walk round, and gives earlier warning. It does not remove the engineer who decides what that warning means.

Never use a model for: protection settings • trip and interlock logic • permit and clearance decisions • statutory returns without verification • anything reaching the plant without an engineer's signature

The vocabulary you will hear this year

Eleven words grouped — plain meaning, plant example, and what to watch for

Terms	Plain meaning	Plant example	Watch out for
Model • Training	A rule learned from history, and the act of learning it	One model per ID fan, trained on a year of healthy running	A model trained on a faulty period learns the fault as normal
Feature • Label • Residual	The input you build • the answer you teach • actual minus expected	Load-normalised bearing rise • lab GCV • six unexplained degrees	No label means no supervised model — establish that on day one
Drift • Inference • Edge	The plant moves away from the model • running it • running it near the machine	Model stale after overhaul • a residual every minute • a box in the LT room	Nobody owns retraining after an overhaul
LLM • RAG • Agent	Language model • the same model grounded in your documents • one that calls tools	Draft a trip report • answer from the O&M manual with a citation • assemble a work order	Ungrounded answers invent numbers; an agent needs an approval gate

If a vendor uses a word you cannot map onto this table, ask them to map it. The good ones manage it in one sentence.

The data pathologies that quietly destroy models

None of these show up on an operator screen. Every one shows up in the model.

What is wrong with the data

- Historian compression and exception deviation set for screen trends, not analysis
- A frozen transmitter reading a perfectly plausible constant value
- Tags on the DCS graphic that were never configured to historise at all
- DCS tag name, historian point and SAP equipment number that do not agree
- Timestamps from clocks never synchronised across DCS, historian and lab
- Coal and chemistry results living in spreadsheets outside any system

What it does to the model

- The slow drift the model exists to catch is discarded before it is ever stored
- The residual goes to zero, and the model reports perfect health on a sick machine
- The signal that would have explained the fault simply does not exist
- Alerts cannot be joined to work orders, so nothing is ever validated
- Cause and effect swap order, and the model learns the relationship backwards
- The answer key for every soft sensor is unavailable to the model that needs it

Not one of these is visible on an operator screen. Every one is fatal to a model. Fix the data before buying the algorithm.

What changed in your head this hour

Block one close

● A threshold asks the wrong question

'Is it above 90 °C' has no context. 'Is it above what it should be at this load and ambient' is the question a model asks.

● The residual is the whole idea

Actual minus expected. Anomaly detection, soft sensing and loss attribution are three uses of one arithmetic operation.

● Trust is earned on unseen data

A model judged on data it has already seen is a sales figure, not an engineering result. Ask what period it was tested on.

● Your data decides the ceiling

Compression settings, frozen transmitters and unhistorised tags cap what any algorithm can do. That work is yours, and it starts before procurement.

Nothing in the next four hours contradicts this hour. It only adds equipment, rupees, and your own hands on the keyboard.

SESSION

From conventional to intelligent monitoring

Why a fixed alarm arrives last, and what a residual sees weeks earlier

- The P-F curve and where each technique sits on it
- Anomaly detection without the mathematics
- The residual pipeline, end to end
- Alert quality, and the alert-to-work-order loop

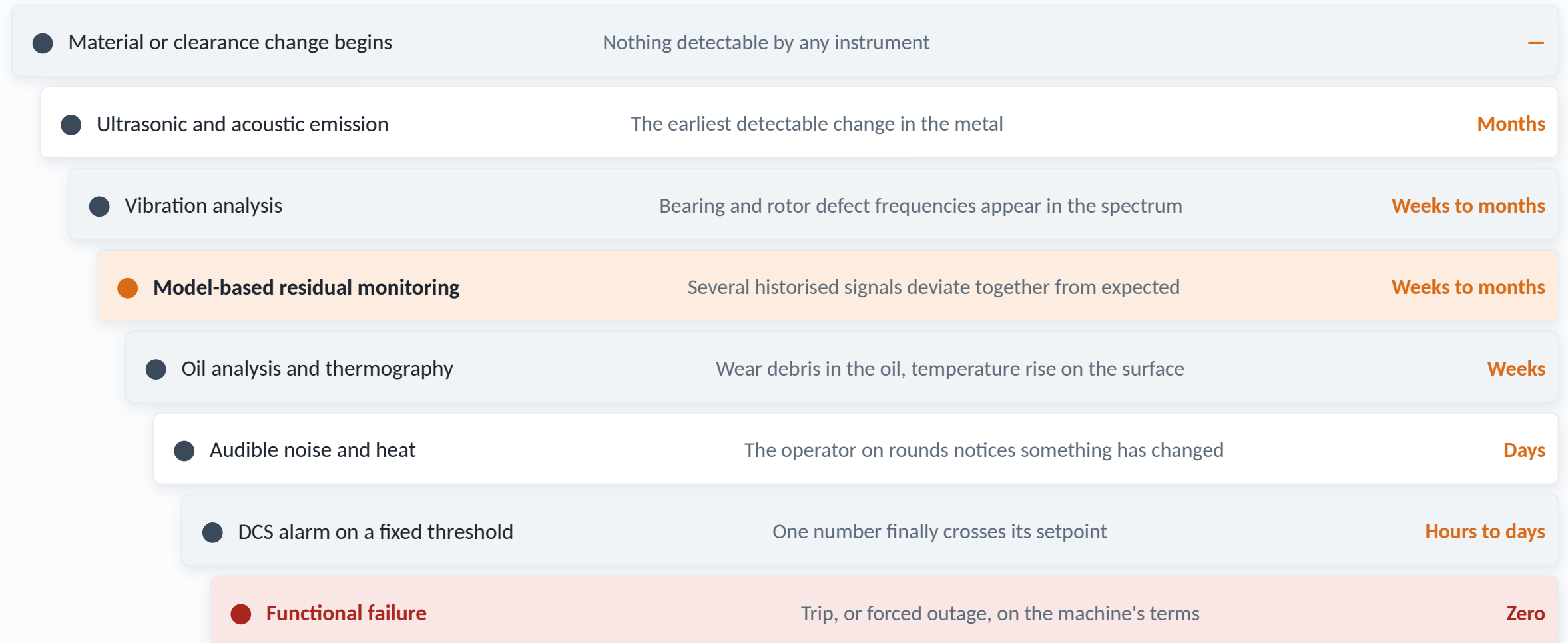
Hands-on: SIM-1 — residual anomaly detection

11:25 – 12:10

45 minutes

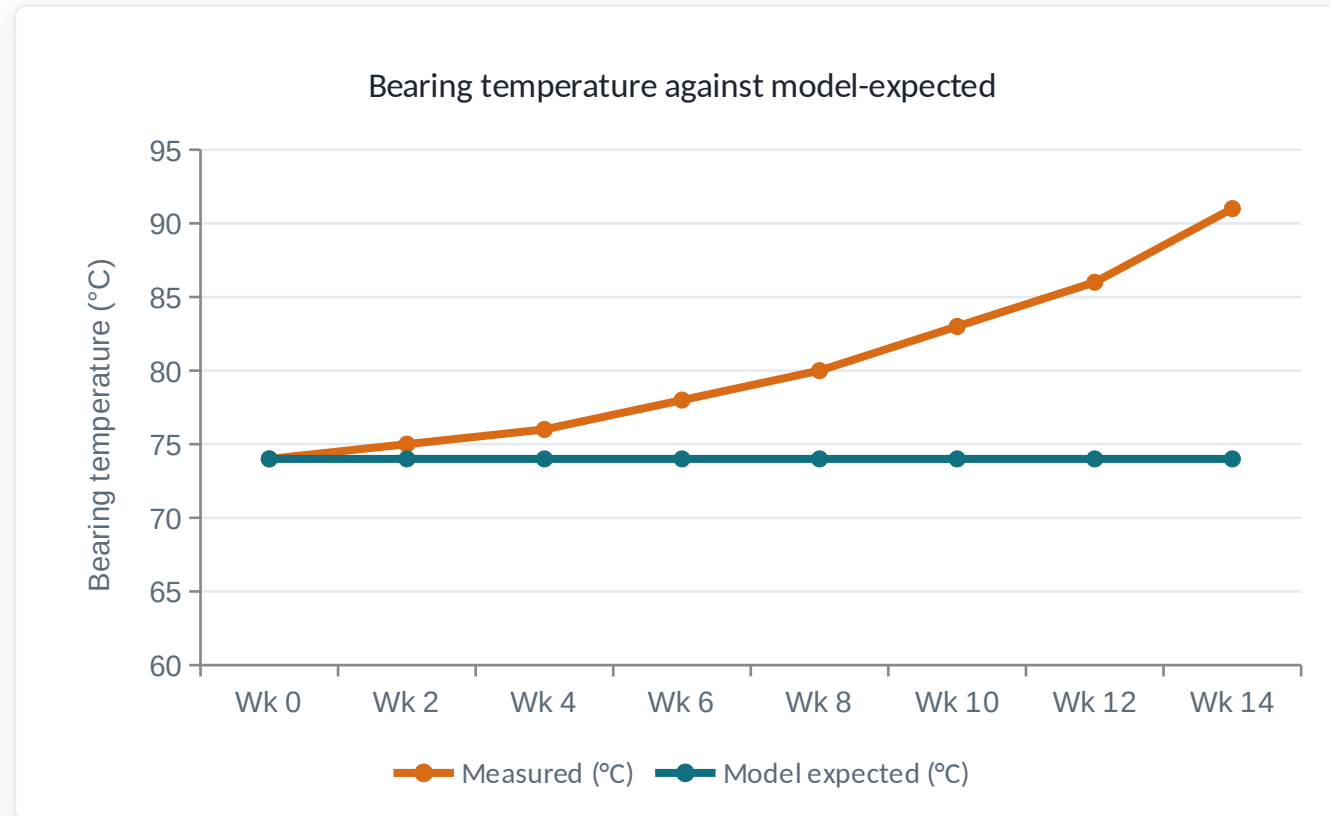
The P-F curve, and where each technique sits

From first detectable symptom to functional failure — who notices, and when



Why a fixed threshold is structurally late

ID fan drive-end bearing — an illustrative reconstruction, not a MAHAGENCO case study. Same machine and the same numbers as handout 2.2.



Week 3 — residual +2 °C. Load and ambient are already removed, so this is the machine and not the weather. Nothing on any screen in the control room has changed.

Week 5 — residual +4 °C, sustained across several load points. The system alerts. The bearing reads 78 °C, which is 12 °C BELOW the alarm and has happened normally hundreds of times.

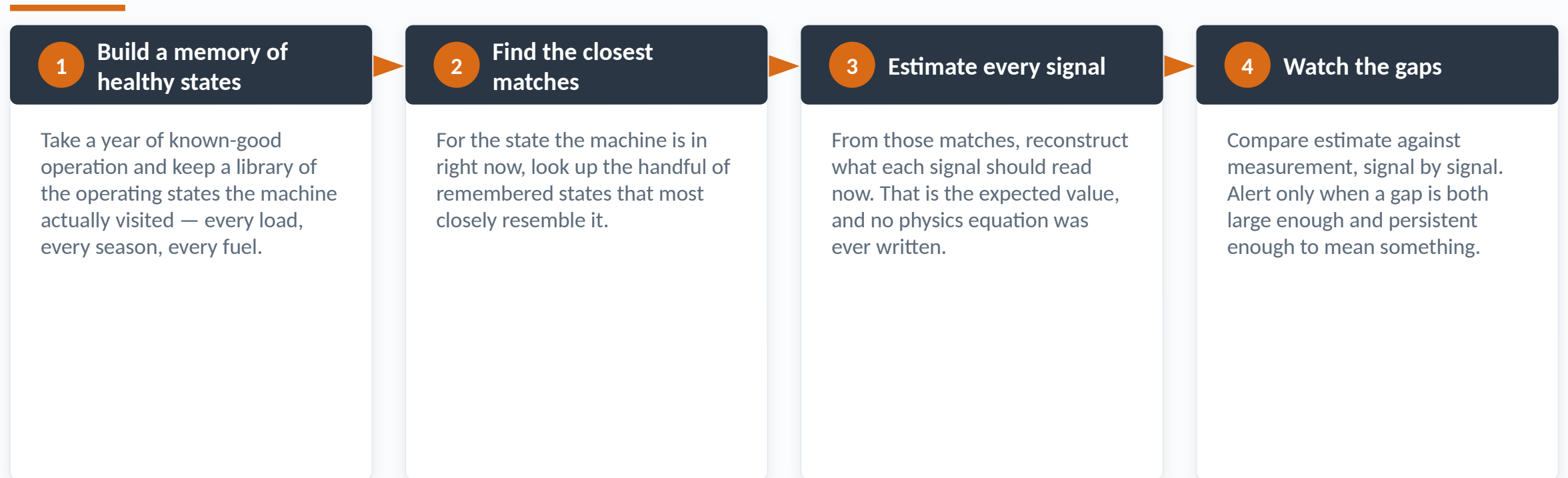
Week 10 — 86 °C on a warm day at full load. An operator finally notices, because now the absolute number looks wrong.

Week 10+ — the DCS alarm at 90 °C. The fan comes out on the fan's terms rather than yours. The alarm was not wrong; a fixed threshold is simply the last thing in the plant to know.

Five weeks of notice against a few days. Why not just lower the alarm to 78? Because 78 is normal at full load in May — you would create a nuisance alarm that is inhibited within a fortnight.

Anomaly detection without the mathematics

What a similarity model actually does, described in plant language



No failure history is required. The model only needs to know what healthy looks like, in every condition the machine actually runs in.

One signal moving, or the machine moving

The single most useful discrimination a multi-signal model hands you

ONE TAG DEVIATES ALONE

That is instrumentation

Bearing temperature rises, but fan power, lube oil outlet, vibration and discharge pressure all read exactly as expected. Nothing physical can raise one and leave the rest untouched. Check the transmitter and the thermocouple before you touch the machine.

A COHERENT GROUP DEVIATES

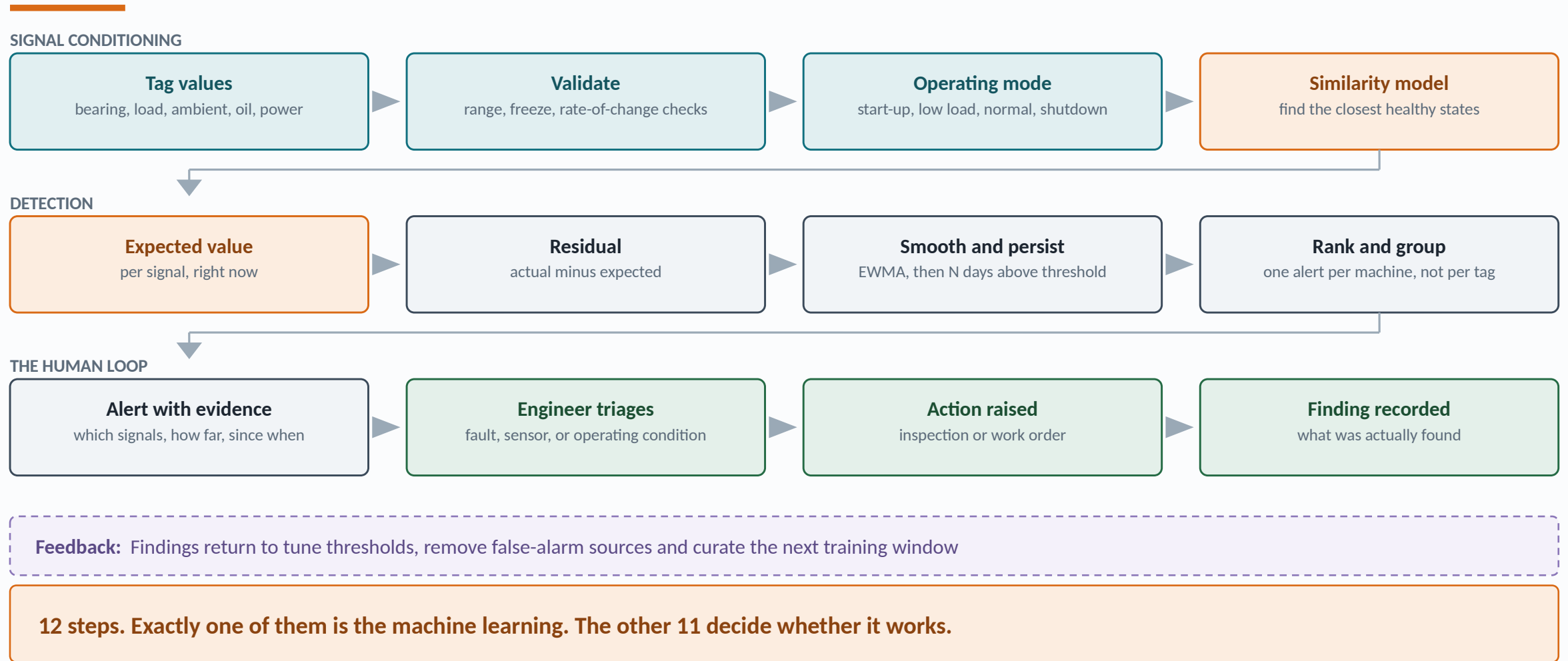
That is the machine

Bearing temperature, lube oil outlet, vibration and motor current all move together in the direction physics predicts. Not one has crossed an alarm limit. The pattern is the evidence, and it appears weeks before any individual number does.

Which of the two it is decides who you send: the instrument technician, or the mechanical engineer with a spare bearing. No single-tag alarm can tell you that.

The residual pipeline, end to end

Everything between a tag in the historian and an engineer picking up the telephone



Operating modes — why one model is never enough

A model trained only on full load will scream through every single start-up

● The problem

Start-up, low load, full load and shutdown are physically different machines. A residual computed against full-load behaviour is meaningless during a ramp.

● Segmentation

Detect the mode first — from load, mill combination and firing state — then apply the model built for that mode. Suppress alerts during transitions you have not modelled.

● Why it is first-order here

Units designed for baseload now two-shift and run at technical minimum. Transitions are no longer rare events; they are a large part of the month.

● The design consequence

Budget for several models per machine and a written rule for which one is live. That is engineering work, not a configuration screen.

If a vendor demonstration only ever shows steady full load, ask to see a start-up. The answer tells you how much work is left.

Lab — SIM-1 · Residual anomaly detection

Twelve minutes at the keyboard, on the ID fan bearing case we have just walked through

SIM-1

Residual anomaly detection — ID fan bearing

12 minutes at your laptop

Do this

1. Open SIM-1 and run the default case. Write down the lead time the residual gives you against the 90 °C fixed alarm.
2. Drag the residual threshold down, step by step. Watch the lead time improve and the false alarm count climb with it.
3. Set persistence to 1 day and re-run. Count how many single-sample spikes now raise an alert.
4. Put persistence back to 3 days and settle on a threshold you would be willing to defend in a morning meeting.
5. Tick 'train on contaminated history' and run again. Watch what happens to the detection.

Watch for

- The residual crosses its threshold weeks before the measured temperature reaches 90 °C.
- Every reduction in threshold buys lead time and costs false alarms. There is no setting that gives you both.
- At one day of persistence, sensor spikes become alerts; at three days, they do not.
- Trained on contaminated history, the model treats the fault as normal and raises nothing at all.

Two lessons, proven rather than asserted: a residual sees the fault weeks before a fixed threshold, and a model trained on history that already contained the fault will never detect it. Both were claims this morning. Now they are yours.

Alert quality — the numbers that decide adoption

Measure the programme the way the shift engineer actually experiences it

Measure	What it means	Healthy	Failing	Who judges it
Lead time	Warning before the conventional alarm or the failure	Weeks	Hours	Maintenance planning
False alarms per model-month	Alerts raised with nothing to find	Below one	Several a week	The engineer who chases them
Precision on actioned alerts	Share of investigated alerts that found something	Most of them	A small minority	Station leadership
Actionability	Does the alert name a signal, a reason and a next step	Always	A number only	The shift engineer
Time to triage	How long an alert sits before somebody looks at it	Same day	Weeks	The named owner
Feedback closure	Findings written back against the alert	Every one	Almost none	The programme owner

₹100.87
crore

cumulative fixed-charge
disallowance for availability
shortfall across the fleet — what
late detection actually costs us

What intelligent monitoring can and cannot detect

Be as precise about the right column as the left, or the first miss ends the programme

It will catch this

Slow degradation with a signature in historised tags

Fouling, wear, leakage, blockage, loss of cooling

A group of signals drifting quietly together

A transmitter that has stopped responding

Behaviour unlike anything in the healthy history

A machine degrading over weeks

It will not catch this

A failure with no measured signal anywhere near it

A sudden mechanical break with no precursor at all

A fault the model was trained through and learned as normal

A transmitter wrong in a physically plausible way since day one

Any operating condition the training year never contained

A machine failing in the next ten seconds — that is the protection system's job

It buys weeks on degradation. It buys nothing on sudden failure, and claiming otherwise is how programmes lose their credibility.

The five ways monitoring quietly stops working

Every one of these leaves the dashboard looking perfectly healthy

Compression

Historian compression and exception deviation tuned for clean screen trends discard exactly the slow drift the model exists to find. The trend still looks fine; the evidence has been thrown away.

Frozen transmitter

A stuck input reads a plausible constant. The residual falls to zero, the model reports perfect health, and the machine degrades unwatched behind it.

Timestamp misalignment

Unsynchronised clocks across DCS, historian and lab shift signals against one another. The model learns cause and effect in the wrong order and its estimates decay.

Post-overhaul drift

After an overhaul the machine is genuinely different. An un-retrained model alarms continuously, is disbelieved, and is switched off inside a month.

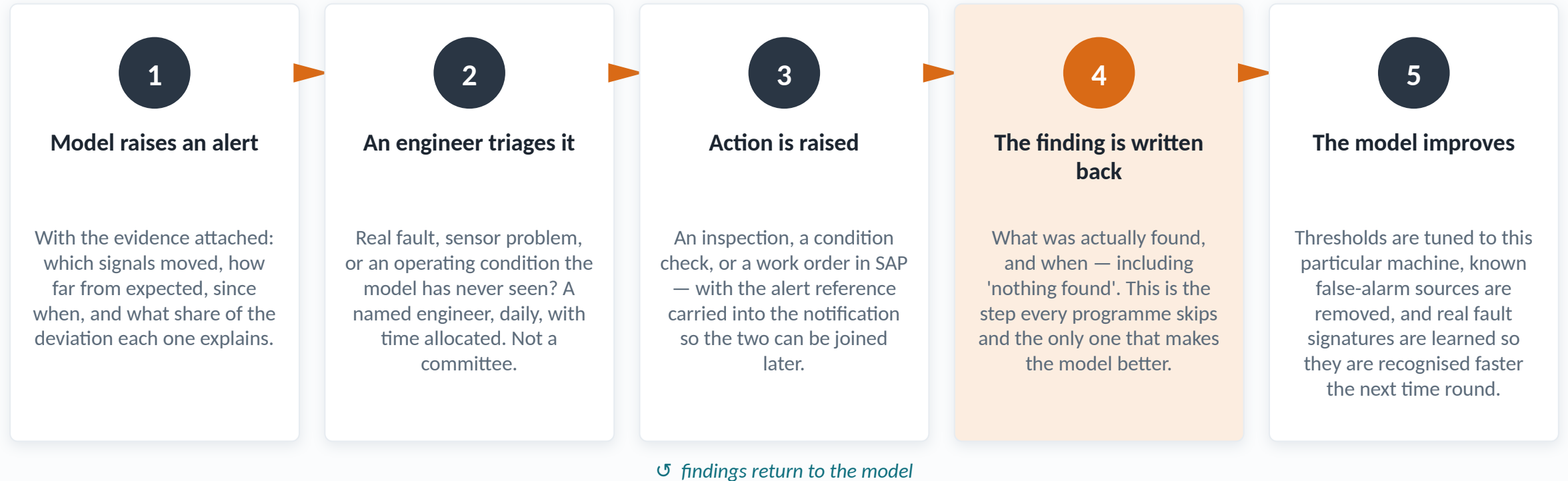
Silent tag loss

A point stops updating after a system change and nobody notices, because no alarm exists anywhere for a tag that has simply gone quiet.

Never assume a quiet model means a healthy machine. Check for: frozen inputs • dead tags • unchanging residuals • models not retrained since the last overhaul

The alert-to-work-order loop

The technology is the easy part. This loop decides whether the programme is alive in a year.



A monitoring programme without step four dies within a year. Every time, at every station, without exception.

Fleet-scale monitoring

One model per machine becomes hundreds of models across five stations

● The arithmetic

One model per fan, mill, pump, motor and heat exchanger. A single unit reaches dozens. Five stations reach several hundred. Indicative — but the order is right.

● Nobody watches hundreds of screens

At fleet scale the output must be one ranked daily list, worst deviation first with the evidence attached — not a dashboard per machine.

● Ownership must be named

Each model needs an owner at the station, and the fleet needs one owner of the method. Unowned models get switched off quietly, within months.

● A model is a configured asset

It needs a register, a version, its training period on record, a retraining trigger and a review date — the same discipline as a protection setting.

● Stations are not interchangeable

Khaperkheda Units 1-4 ran at 57.85 % availability in June 2026, Parli Unit 8 at 97.40 %. The same model does not transfer between them.

● Governance beats algorithms

Fleet programmes fail on registers, ownership and retraining — not on mathematics. Budget for the administration, or do not start.

Hundreds of models is a governance problem wearing an engineering costume. Decide who owns them before you buy them.

Three things to take to lunch

Block two close

- **A residual sees weeks; a threshold sees days**

You proved it at the keyboard in SIM-1. The gap between the two is the entire business case for intelligent monitoring.

- **A model is only as honest as its training period**

Contaminate the healthy history and detection collapses completely. You did not take that on trust — you watched it happen.

- **Without the feedback step, the programme dies**

Alerts that are never closed out teach the model nothing, and teach the engineer to stop opening them.

- **After lunch**

We put rupees on it — heat-rate loss attribution and auxiliary power, worked on MAHAGENCO's own June 2026 numbers.

₹100.87 crore of fixed-charge disallowance and ₹56 crore of heat-rate gap in a single month are waiting on the other side of lunch.

SESSION

AI for predictive maintenance

Boiler feed pumps, mills, fans, transformers and motors — and the availability money behind them

- Equipment by equipment: what is detectable and how early
- Remaining useful life, and the question a planner actually asks
- What predictive maintenance needs from the organisation
- ₹100.87 crore of availability disallowance, and where it came from

12:10 – 13:00

50 minutes

Hands-on: SIM-5 — predictive maintenance and RUL

Maintenance strategies compared

Four positions on one ladder — and most stations are two rungs lower than they believe

Strategy	What triggers the work	What it costs you	Where it fits
Breakdown — run to failure	The machine stops	Collateral damage, long outage, no spare staged	Cheap, duplicated, non-critical items only
Time-based — calendar or running hours	The date arrives	Good machines opened; bad ones fail before the date	Statutory items and known wear-out parts
Condition-based — measure, then decide	A measured parameter crosses a limit	Route data monthly; the fault can start the day after	Vibration routes, oil analysis, thermography
Predictive, then prescriptive	A model projects a limit will be crossed	Data plumbing, a triage engineer, and discipline	Rotating plant with continuous data

Prescriptive is the next rung: not only when it will fail, but which action costs least. Nobody reaches it from row one.

Case — boiler feed pumps

Motor-driven on the 210 and 250 MW units, turbine-driven on the 500 and 660 MW machines

What the model detects

- Bearing degradation — thrust and journal, weeks ahead of temperature alarm
- Mechanical seal wear, seen in leak-off flow and temperature
- Cavitation and loss of NPSH margin at low suction pressure
- Recirculation valve passing, sticking or opening late
- Hydraulic performance drift — head developed falling at the same speed and flow
- Coupling wear and shaft alignment deterioration
- Balance drum and balancing disc wear, seen in axial position and leak-off

Signals it uses

- Suction and discharge pressure, differential head, feed flow
- Motor current and power, or turbine steam flow and speed on a TDBFP
- Bearing metal and oil temperatures, shaft vibration and axial position
- Seal leak-off flow, temperature and seal water differential pressure
- Recirculation valve position and demand, and its response time
- Deaerator level, suction temperature and calculated NPSH available

Typical lead time

Bearing and seal problems typically show in the residual weeks before a temperature alarm; hydraulic drift shows in months and is almost never noticed at all.

Proven — the most reliable predictive application on the feedwater side

Watch out: Do not use one model for both drive types. A turbine-driven pump on a 660 MW unit has speed as a free variable; a motor-driven pump does not. Their normal behaviour is genuinely different and one model will fit neither.

Case — coal mills and pulverisers

The plant item that most often decides whether a unit holds load

What the model detects

- Grinding element and roller wear, trended as current per tonne of coal
- Mill choking, hours before it happens, from differential pressure behaviour
- Fineness deterioration between sampling campaigns
- Feeder calibration drift — the mill says one tonnage, the boiler says another
- Classifier setting problems and seal air differential loss
- Mill fire precursors — outlet temperature rise with falling coal flow

Signals it uses

- Mill motor current and power, and the ratio to feeder tonnage
- Mill differential pressure and primary air flow and temperature
- Mill outlet temperature and the cold-air damper demand
- Grinding or hydraulic loading pressure and roller position
- Seal air to mill differential pressure
- Reject rate, mill vibration and bowl or table speed

Typical lead time

Choking is predictable hours ahead from differential pressure and current behaviour. Grinding element wear trends over months and tells you what to stage for the next outage.

Proven — high value, because a mill trip is a load restriction

Watch out: Build one model per mill, never one per unit. Mills on the same boiler have different wear, different coal splits and different air balance. A unit-level model averages six machines and hides the one that is degrading.

Case — ID, FD and PA fans

Large rotating plant, continuously instrumented, and directly on the auxiliary power bill

What the model detects

- Blade and impeller erosion from ash, seen as efficiency drift
- Unbalance developing from uneven erosion or ash build-up
- Bearing wear on both fan and motor, ahead of temperature alarm
- Damper, inlet guide vane and blade-pitch actuator problems
- Approach to surge or stall on axial fans, before it is audible
- Duct and expansion joint leakage, seen as flow that does not arrive
- Fan efficiency drift — power rising for the same gas flow

Signals it uses

- Motor current and power, and fan speed where variable
- Inlet and outlet pressure, differential head across the fan
- Gas or air flow, and furnace draught
- Bearing vibration, shaft displacement and bearing temperatures
- Damper, IGV or blade-pitch position demand and feedback
- APH inlet and outlet gas temperature and oxygen, for leakage context

Typical lead time

Bearing and unbalance faults appear in the residual weeks ahead. Efficiency drift from erosion is a months-long trend and is the one that costs money continuously.

Proven — and the easiest place to connect maintenance to auxiliary power

Watch out: Fan power against gas flow is the cheapest health indicator in the plant, needs no new instrument and no model, and almost nobody plots it. Ask your performance engineer for that curve for last year before you buy anything.

Case — transformers and switchyard

Identical equipment at all five stations, and the highest consequence per failure

What the model detects

- Dissolved gas trends and ratios, not single values against a limit
- Winding hot spot running above what the thermal model predicts
- OLTC contact wear, from motor current signature during a tap change
- Bushing capacitance and tan delta drift, well before failure
- Cooling bank underperformance — fans or pumps running with no effect
- Moisture ingress and accelerated paper ageing at sustained high load

Signals it uses

- Dissolved gas concentrations and their rate of change over months
- Top oil and winding temperatures, load current and ambient
- Cooler bank status, fan and pump currents, oil flow indication
- OLTC motor current profile, operation counter and drive time
- Bushing leakage current or online tan delta where fitted
- Partial discharge, where online monitoring is installed

Typical lead time

Gas trends and bushing drift develop over months. The lead time is long, which is exactly why the sampling interval, not the algorithm, is usually the limiting factor.

Proven — dissolved gas trending is the oldest predictive technique in the industry

Watch out: Six-monthly manual sampling cannot produce a trend. Two points a year tell you almost nothing about a rate of change, and rate of change is the whole diagnosis. Fix the sampling interval before anybody discusses an algorithm.

Case — HT and LT motors

Hundreds per unit, and the electrical signal itself is the sensor

What the model detects

- Broken or cracked rotor bars and end-ring faults, from current sidebands
- Static and dynamic eccentricity in the air gap
- Bearing defect frequencies, visible in current as well as vibration
- Stator insulation deterioration, from partial discharge and unbalance trends
- Supply quality — voltage unbalance, harmonics and their heating effect
- Driven-equipment problems seen through the motor, without instrumenting the machine
- Cooling degradation — blocked coolers and rising winding temperature at constant load

Signals it uses

- Stator current waveform at high sample rate, not an averaged value
- Supply voltage, unbalance and harmonic content
- Winding and bearing temperatures, and cooling air or water temperature
- Vibration where fitted, for cross-confirmation of a current-based finding
- Load context — driven equipment flow, pressure or tonnage
- Start counts, run hours and starting current profile

Typical lead time

Rotor bar and eccentricity faults are detectable months ahead. The technique is mature; the obstacle is almost always the data path, not the mathematics.

Proven — motor current signature analysis is decades old and well validated

Watch out: Historian averages destroy this. The sidebands MCSA depends on sit close to the supply frequency and vanish the moment current is stored as a one-minute average. This case needs edge processing near the machine, not a historian tag.

Case — ESP, ash handling and CHP conveyors

A large share of the maintenance effort, and almost none of the instrumentation

What the model detects

- ESP field energisation drift and rapping that has stopped being effective
- Hopper ash build-up and choking, predicted from evacuation cycle times
- Ash slurry and vacuum pump wear, from current and cycle duration
- Conveying line blockages, from vessel fill and empty times lengthening
- Conveyor belt misalignment, idler seizure and drive slip
- Transfer chute and crusher choking in the coal handling plant
- Wagon tippler and stacker-reclaimer mechanism degradation

Signals it uses

- ESP field kV, mA and spark rate, per field, and stack opacity
- Ash vessel fill and evacuation cycle times, and valve operation counts
- Conveyor motor current, belt drift switch operations and pull-cord events
- Bearing temperatures and vibration on drive and tail pulleys
- Crusher and ring granulator current, and screening throughput
- Cycle counts and run hours, which most stations already log somewhere

Typical lead time

Cycle-time drift is the earliest and cheapest indicator here, and it is usually already recorded in the PLC without anybody trending it.

Emerging in practice — the technique is simple, the instrumentation is the gap

Watch out: This plant consumes a large share of the station's maintenance effort and is rarely instrumented for analytics at all. The first project here is not a model. It is getting cycle times and motor currents into the historian.

Diagnostics versus prognostics

Two different questions, two different levels of confidence, two different audiences

DIAGNOSTICS

What has changed?

The residual on the thrust bearing is six degrees and growing, and the model points at that signal rather than at load or ambient. It names the machine, the signal and often the direction. It is a question about the present, and an engineer can go and check it.

PROGNOSTICS

How long have I got?

The same residual, fitted with a degradation curve and projected forward to a functional failure threshold. It answers a planning question rather than a diagnostic one, and it arrives with an uncertainty band that is usually wide. Wide is not useless. Wide is honest.

Diagnostics tells you to look. Prognostics tells you whether to book the outage. Most stations need the first far more than they currently think they need the second.

Remaining useful life, and why a point estimate is the wrong answer

The number a planner can actually act on is a probability, not a date

● Sixty-two days is not an answer

A single number invites either complacency or panic. No planner commits an outage window to it, and no engineer defends it in a review.

● The band carries the information

A projection with a ninety per cent range tells you what you can commit to. The width of that band is the message, not an apology for it.

● Ask the planner's question instead

What is the probability this machine survives to the outage already in the plan? That question has a usable answer and a rupee value attached.

● What makes the band open

A short trend history, a noisy measurement, and a failure threshold nobody has ever measured. The first of those is the cheapest to fix.

● More history beats a better algorithm

Thirty days of trend cannot separate fast degradation from slow degradation with noise on it. No algorithm recovers what was never observed.

● Statistical error is not model error

The band covers uncertainty in fitting the curve. It says nothing about the curve being the wrong shape, which is usually the larger risk.

Remaining useful life is a distribution, not a date. Anyone quoting a single number has already hidden the part you needed.

Lab — SIM-5 · Predictive maintenance and remaining useful life

Ten minutes at the keyboard, on the station carrying the largest disallowance in the fleet

SIM-5

Predictive maintenance and remaining useful life

10 minutes at your laptop

Do this

1. Open SIM-5 and select Koradi Units 8-10 — the station carrying ₹28.04 crore of disallowed fixed charge.
2. Read two figures only: the probability of surviving to the planned outage, and the expected cost of waiting against the cost of acting now.
3. Drag 'days of trend observed' down to 30 and watch the projection band open up. Do not change anything else.
4. Put it back to 70, then move the planned outage date further out and closer in, and watch the recommendation flip.
5. Set the forced-outage duration to a figure your own station would recognise, and see what that does to the asymmetry.
6. Press 'New machine' twice. The same settings give a different answer, because the trend is a different sample.

Watch for

- The remaining-life figure moves very little when you shorten the trend, but the ninety per cent range widens dramatically.
- The decision table is driven by the survival probability, not by the point estimate of remaining life.
- Moving the planned outage flips the recommendation with no change to the machine or its degradation at all.
- The cost of a forced outage carries a share of the fixed-charge disallowance the station is already paying.

The point estimate barely moves; the uncertainty band moves enormously. That is the lesson. A planner acts on the probability of reaching the outage already scheduled, and that number is worth more than any single date.

The availability arithmetic

What unavailability already cost us — June 2026, from our own bill

Station	Availability %	Normative %	AFC disallowance ₹ cr	Reading
Koradi Units 8-10	72.50	85.00	28.04	Largest single loss in the fleet
Chandrapur Units 3-7	64.88	80.00	23.94	Lower norm, still far short
Khaperkheda Units 1-4	57.85	85.00	21.75	Lowest availability in the fleet
Chandrapur Units 8-9	74.87	85.00	11.75	Newer units, same shortfall
Bhusawal Unit 6	79.01	85.00	7.88	Six points from the norm
Paras Units 3-4	74.83	85.00	5.74	Ten points from the norm
Koradi Unit 6	75.46	75.00	0.25	Norm is 75 % — essentially met
Parli Unit 8	97.40	85.00	0.00	Fleet best — the norm is achievable

₹100.87
crore

cumulative fixed-charge
disallowance for availability
shortfall, of which ₹32.93 crore
was adjusted in June alone

What predictive maintenance needs from the organisation

The technical part is the smaller half — this is where programmes are actually won

The alert-to-work-order loop

- One named engineer owns triage, with a defined response time per criticality
- Every alert carries its evidence — the signals, the residual and the trend — into the notification
- A model alert becomes a CMMS notification, not an email that dies in an inbox
- Every alert is closed with a finding: confirmed, not confirmed, or not inspected
- Findings go back to the model owner monthly — that is the only way accuracy improves
- Alert quality is reviewed as a number, the same way trip statistics are

What has to change around it

- Spares policy — condition-based means the part is staged before the outage, not ordered after
- Outage scope is built partly from model output, and something must come off the list in exchange
- Instrumentation gaps are funded as part of the programme, not deferred to next year
- Two trained people per station, so the programme survives a transfer
- Contract and warranty terms updated where a vendor is doing the monitoring
- A written rule that no model output enters protection or interlock logic

An alert that does not become a work order has cost you money and earned nothing. The loop is the programme.

Why predictive maintenance programmes fail

Eight failures seen repeatedly across the industry, and what prevents each one

WHY IT FAILED

Trained on history that contained the fault

Alerts went to an inbox nobody owned

Too many alerts in the first fortnight

No feedback on what the alert found

Transmitter replaced, model never retrained

Model output could not reach the work order

Pilot proved value, nobody funded scale-up

The one engineer who understood it moved

WHAT PREVENTS IT

Check the training window against CMMS work orders first

One named triage engineer and a response time in writing

Start with three machines and tune persistence before more

Close each alert: confirmed, not confirmed, or not inspected

Instrument changes raise a model-review task automatically

Build the historian-to-CMMS link inside the pilot, not after

Agree the scale-up trigger and budget before the pilot

Two trained people per station, plus written documentation

Not one of these eight is a modelling failure. Every one of them is an organisational failure, and every one is preventable.

Block close — four things to carry into lunch

Everything after the break is about our own numbers rather than equipment

● The residual, and then the trend

Monitoring says something has changed. Predictive maintenance fits a curve to that change and turns it into a date with a band around it.

● One model per machine, never per unit

Per mill, per fan, per pump. A unit-level model averages six machines and hides precisely the one that is degrading.

● An alert is not a work order

Nothing in this block earns a rupee until a named engineer triages it and the evidence reaches the CMMS with the notification.

● ₹100.87 crore

The cumulative fixed-charge disallowance for availability shortfall. That is what unavailability has already cost, before anybody buys anything.

After lunch we stop discussing equipment and start reading our own bill. Bring the same scepticism.

4

SESSION • THE PIVOT OF THE DAY

Reading our own numbers

MAHAGENCO's June 2026 regulatory filing, turned into an opportunity map

- The decomposition: auxiliary power against boiler and turbine
- What our fuel actually costs us, station by station
- The merit order as a scoreboard
- Where the ₹56 crore a month actually sits

Hands-on: SIM-2 — heat rate loss attribution

14:30 – 15:00

30 minutes

Where this data comes from

Three documents this company already produces every month, and one that nobody reads

● The June 2026 energy bill

Bills 001/2026 and 35/2026. Station by station availability, PLF, gross and net generation. 4,588.47 MU sent out, ₹2,806.55 crore net.

● The Part-I FSA bill and its F10 sheet

Provisional bill 43/2026, with the F10 energy charge rate calculation. Heat rate, auxiliary consumption, GCV, oil and transit loss, per station.

● The MERC merit order stack

July 2026 revision zero, effective 16.07.2026 to 15.08.2026. Every station's variable charge and where it sits in the dispatch queue.

● The document nobody reads

F10 is prepared for billing and then filed. It is the most complete performance record in the company, and it is not a performance report to anyone.

Nothing in this block is a benchmark, a vendor study or an assumption. It is our own filing, read as an engineering document.

The fleet in one table, June 2026

Availability, PLF and net heat rate against MERC norm — eight groups chosen to show the spread

Station	Avail %	PLF %	Net HR act / norm	Gap kcal/kWh
Nashik Units 3-5	94.62	58.40	2,804 / 2,754	+50
Parli Unit 8	97.40	79.55	2,461 / 2,415	+46
Khaperkheda Unit 5	87.86	82.03	2,385 / 2,375	+10
Bhusawal Unit 6	79.01	70.37	2,183 / 2,139	+44
Paras Units 3-4	74.83	66.92	2,578 / 2,415	+163
Koradi Units 8-10	72.50	66.62	2,442 / 2,230	+212
Chandrapur Units 3-7	64.88	53.56	2,709 / 2,688	+21
Khaperkheda Units 1-4	57.85	54.55	2,715 / 2,630	+85

₹56 crore

the net heat-rate gap against
MERC norms across thirteen
thermal groups, in June 2026
alone

The decomposition that changes the conversation

One line of arithmetic that tells you whose problem the heat-rate gap actually is

THE IDENTITY

$$\text{Gross HR} = \text{net HR} \times (1 - \text{aux fraction})$$

Every station already reports a net heat rate and an auxiliary consumption percentage. Multiply one by one minus the other and you have the gross heat rate — what the boiler and turbine actually achieved. MERC publishes a norm for both. Almost nobody in the plant does this arithmetic.

WHAT IT REVEALS

A combustion problem, or an electrical one

The net gap splits into two pieces with two different owners and two entirely different remedies. A positive gross gap is combustion, condenser and air pre-heater. A negative gross gap means the plant is beating its norm and every kilocalorie of the penalty is electricity consumed inside the fence.

Nashik's gross heat rate is 2,440 against a normative 2,458 — the boiler and turbine are beating the norm by 18 kcal/kWh, and the entire 50 kcal/kWh penalty is auxiliary power at 12.96 % against 10.75 %.

Station by station — where the gap actually sits

The same net gap, decomposed. A negative gross gap means the plant is beating its norm.

Station	Aux gap pp	Gross HR act / norm	Gross gap	Diagnosis
Chandrapur Units 3-7	+3.02	2,393 / 2,455	-62	Auxiliary power is the whole problem
Nashik Units 3-5	+2.21	2,440 / 2,458	-18	Auxiliary power is the whole problem
Parli Unit 8	+2.24	2,197 / 2,210	-13	Auxiliary power is the whole problem
Khaperkheda Unit 5	+0.57	2,228 / 2,232	-5	Auxiliary power is the whole problem
Khaperkheda Units 1-4	+2.74	2,377 / 2,375	+2	Both, roughly balanced
Koradi Unit 6	+0.56	2,246 / 2,190	+56	Mostly boiler and turbine
Paras Units 3-4	+2.29	2,279 / 2,190	+89	Mostly boiler and turbine
Koradi Units 8-10	+0.95	2,272 / 2,096	+176	Mostly boiler and turbine

₹22.84

Koradi 8-10's GROSS gap: 176 kcal/kWh on 964.8 MU generated. Its NET gap of 212 on 897.75 MU sent out is worth ₹25.61 crore. Same station, two questions — and mixing them is the error this whole session is

about

What our fuel actually costs us

Cost of heat = MOD variable charge ÷ net heat rate. The station's own money, not an assumption.

Station	MOD variable charge ₹/kWh	Net heat rate kcal/kWh	Cost of heat ₹/kcal	Implied as-fired coal ₹/tonne
Nashik Units 3-5	5.9800	2,804	0.002133	6,217
Bhusawal Unit 3	4.9470	2,848	0.001737	5,280
Koradi Units 8-10	3.2840	2,442	0.001345	4,116
Khaperkheda Unit 5	3.1890	2,385	0.001337	3,655

70 %

spread in implied as-fired coal cost between Khaperkheda Unit 5 and Nashik, inside one company

Indicative basis

Cost of heat = each station's own MERC merit-order variable charge divided by its own actual net heat rate for June 2026. Implied coal cost = cost of heat × as-fired GCV. These are the station's own figures, not an assumption.

Nashik's coal costs ₹6,217 a tonne against ₹3,655 at Khaperkheda 5 — nearly twice. The same kcal/kWh saved is worth far more at Nashik.

The merit order is a scoreboard

How heat rate and coal cost decide whether a station that is ready actually gets called

● ₹5.98 per kWh

Nashik's July 2026 variable charge. The most expensive coal station in the MSEDCL stack, ranked tenth overall and behind only gas.

● 94.62 per cent available

The highest availability in the fleet. The plant is ready, declared and staffed. Availability is not the constraint here.

● 58.40 per cent PLF

And it is not being called. The gap between those two numbers is the merit order doing exactly what it is designed to do.

● The causal chain

Heat rate and coal cost set the variable charge. The variable charge sets the rank. The rank sets the PLF. The PLF sets fixed-cost recovery.

● Where the lever is

Rank is relative and nobody controls the other bidders. The only lever inside the fence is the station's own variable charge.

● This is structural, not blame

1979 to 1981 vintage units on the most expensive coal in the fleet. The mechanism is arithmetic, and it is worth understanding rather than resenting.

Every kilocalorie per kilowatt hour and every rupee per tonne moves you up or down a ranked list that decides whether you run at all.

The auxiliary power bill

Electricity generated and then consumed inside the fence, above the approved norm

Station	Aux actual %	Aux norm %	Excess MU/month	Value ₹ cr/month
Chandrapur Units 3-7	11.69	8.67	23.03	9.52
Nashik Units 3-5	12.96	10.75	5.85	3.50
Khaperkheda Units 1-4	12.44	9.70	9.38	3.42
Parli Units 6-7	12.41	9.30	7.86	3.25
Koradi Units 8-10	6.95	6.00	9.17	3.01
Bhusawal Units 4-5	7.06	6.00	5.85	2.37
Paras Units 3-4	11.59	9.30	5.61	2.22
Parli Unit 8	10.74	8.50	3.26	1.35

81.5 MU

above norm in one month, worth
₹32.9 crore — a component of
the ₹56 crore, never an addition
to it

The oil and the coal

Two costs that sit outside the heat-rate arithmetic and are measured every single day

Secondary fuel oil — burning oil to hold a flame

- Chandrapur Units 3-7: 3.23 ml/kWh against a normative 1.00
- Parli Units 6-7: 2.87 ml/kWh against a normative 0.50 — nearly six times
- Oil support is a symptom of low-load running and flame-stability anxiety
- It is the most expensive heat in the station, by a wide margin
- Much of it is precautionary rather than necessary, and nobody measures which

Coal quality — what arrives is not what was bought

- GCV loss between loading and unloading: 619 to 941 kcal/kg
- As-fired GCV across the fleet: 2,733 to 3,270 kcal/kg
- Paras transit loss 1.924 % against a normative 0.800 %
- Stacking loss taken at 85 kcal/kg at every station
- Six stations blend imported coal at 4,624 to 4,709 kcal/kg as fired

Both columns are measured daily and analysed monthly at best. Both are prediction problems long before they are procurement problems.

Lab — SIM-2 · Heat rate loss attribution

Ten minutes at the keyboard, on your own station's June 2026 numbers

SIM-2

Heat rate loss attribution

10 minutes at your laptop

Do this

1. Open SIM-2 and select your own station from the dropdown. Read the facts panel underneath it.
2. Read the split: how much of your net heat-rate gap is auxiliary power, and how much is boiler and turbine.
3. Select Nashik Units 3-5, then Koradi Units 8-10. Same question, opposite answers.
4. Go back to your own station and move the controllable-parameter sliders — excess oxygen, main steam temperature, back pressure, unburnt carbon, air pre-heater leakage.
5. Drag the auxiliary power reduction slider and see what closing half a percentage point is worth at your own variable charge.
6. Read the red warning at the bottom of the module before you close it.

Watch for

- For Nashik the boiler and turbine bar sits below the line — better than the norm. The whole gap is the auxiliary bar.
- Koradi Units 8-10 are the mirror image: a 176 kcal/kWh plant gap and almost no auxiliary gap at all.
- The fleet chart shows four station groups with negative bars — four whose plant is beating its own norm.
- Every slider move changes the monthly rupee figure immediately, priced at that station's own variable charge.

Two stations, the same penalty on paper, and opposite remedies. The decomposition decides which project is worth funding, and it takes one line of arithmetic on data the commercial department already files every month.

The opportunity map

Five pain points from our own June 2026 filing, each with its evidence and an honest confidence rating

1 • Availability — ₹100.87 cr disallowed • worst Koradi 8-10	Predictive maintenance and early-warning anomaly detection on the plant driving forced outages; outage scope built from model output	Proven — anomaly detection is the safest first project in any station
2 • Auxiliary power — 81.5 MU, ₹32.9 cr • worst Chandrapur 3-7	Combinatorial optimisation of mill, fan and CW pump combinations plus ESP field energisation, offered as operator advice	Proven as advisory — the gain depends on how well you already schedule
3 • Boiler and turbine — 176 kcal/kWh • worst Koradi 8-10	Combustion optimisation with online loss attribution, plus condenser and air pre-heater analytics feeding a daily loss report	Emerging — proven as advice, closed loop needs full engineering controls
4 • Secondary fuel oil — 3.23 ml/kWh • worst Chandrapur 3-7	Flame-stability prediction to avoid precautionary oil support at low load, and start-up sequence optimisation	Emerging — depends on flame scanner and mill data you may not historise
5 • Coal quality — GCV loss 619 to 941 kcal/kg • worst Paras	GCV soft sensor trained on your own laboratory results, rake-level transit-loss analytics and blend optimisation	Proven as a soft sensor — but the error must be reported honestly

These are the values of closing each gap entirely against norm. No programme closes a gap entirely. Fifteen to twenty-five per cent of the identified gap is a defensible first-year planning assumption.

SESSION

What other generators have done

NTPC, Tata Power, Coal India, SAIL — and Vistra and Duke Energy internationally

- Indian public-sector and private deployments, named
- Vistra: 67 units, 1 per cent, US\$ 23 million
- Duke Energy: the operating model worth copying
- Reading a vendor claim without being taken in

15:00 – 15:12

12 minutes

What other generators have actually done

Named companies, named systems — and where the numbers are verified against where they are vendor claims

● NTPC — NePPS

NETRA e-Power Plant Solutions: an AI-based operator advisory system predicting equipment faults from sensor data, developed at NTPC's NETRA research centre and Advanced Computing Center.

● Tata Power — enterprise data platform

Announced April 2026 with Databricks: a unified data and AI platform across a 16 GW+ portfolio. Grid management, renewable forecasting and a 'talk-to-data' agent. Note honestly: thermal generation is not named in the announcement.

● Adani Electricity — Mumbai distribution

AspenTech deployment supporting reliable supply to about three million customers. Distribution, not generation — but the same asset-analytics family.

● Coal India — the fuel end

AI satellite dashboard with ISRO's National Remote Sensing Centre for underground coal fires at Jharia and Dhanbad; over 500 AI surveillance cameras; a 'Digi Coal' platform with 17 AI use cases.

● SAIL Bokaro — adjacent heavy industry

AI-assisted stoichiometric optimisation reported at ₹3.23 crore annual savings and 1,500 tonnes of CO₂ avoided. A combustion-chemistry optimisation, in a PSU, in India.

● NLC India and NPCIL

Memoranda of understanding with IIT Kanpur and IIT Bombay respectively for AI and machine learning capability building. The academic-partnership route, which a state utility can copy directly.

Every one of these is a public-sector or Indian private generator. None of this is a Western experiment.

Vistra — the case that maps onto our problem exactly

A US thermal generator, a heat-rate optimiser, and a fleet rollout with published numbers

Stage	What was done	Result	Note
Pilot — Martin Lake, Texas	Multi-layered neural network heat-rate optimiser recommending setpoints every 30 minutes, built by McKinsey and QuantumBlack	2 % heat rate improvement in three months	Model prediction accuracy reported at 99 % or better
Pilot value	Same plant, annualised	US\$ 4.5 million a year; 340,000 tonnes of CO ₂ abated	One plant
Fleet rollout	67 generating units across 26 plants	1 % average efficiency improvement	The fleet average is half the pilot — that is the honest pattern
Fleet value	Company-wide	US\$ 23 million; 1.6 million tonnes of CO ₂ avoided	Published by the company and its consultants

**1 % → 24
kcal/kWh**

the fleet-average result applied to
Koradi 8-10's actual 2,442
kcal/kWh — against a gap of 212

Duke Energy — the operating model worth copying

Not the algorithm. The monitoring and diagnostics centre around it.

Element	Duke Energy	What it means for a five-station fleet
Coverage	Over 87 % of the generating fleet — about 58,000 MW across seven states	Fleet-wide, not one flagship unit
Scale of models	Over 11,000 models drawing on more than 500,000 data points	One model per machine, not one model per plant
The team	A central monitoring and diagnostics centre staffed by five analysts	Five people watching a fleet — the economics only work centrally
The result	A single early catch reported as avoiding over US\$ 34 million in 2016	One catch paid for the programme many times over

Their competitive advantage is not the software. It is that five named people look at the alerts every single day.

Adani Power — the Indian case, and what it does not tell us

AVEVA World, Paris, October 2024 — their own slides, their own screenshots

What they showed	The detail	Evidence
Scope	AVEVA PI System across 8 thermal locations, 25 units, about 150,000 tags, one central operations centre at Ahmedabad. Udupi (1,200 MW) is one of the eight.	Their own conference slides
Boiler and turbine efficiency	Every loss itemised on screen on a 15-minute AND a 24-hour window. Live screenshot: boiler efficiency 86.72 % last 15 min against 86.49 % last 24 h; turbine heat rate 1,883 against 1,826 design.	Screenshots, dated
Asset Health Index	Four tiers — station, asset, equipment, sensor. Pumps 69, fans 71, mills 77. Individual BFP, CEP and CWP scored. Motor winding and motor bearing temperature are scored inputs.	Screenshots, dated
Is it AI?	The 15-minute efficiency is a thermodynamic enthalpy–entropy module in PI Asset Framework, with coal quality typed in by hand. The AI/ML is Python health scoring. Predictive analytics was still on the roadmap.	Stated on their slides
The published benefit	None. Not a kcal/kWh, not a percentage point of availability, not a rupee. Their own annual report quantifies heat-rate gains only for air-preheater baskets and water-box cleaning — the mechanical work, not the software.	Nothing found

Well-documented capability. Undocumented benefit. Learn to tell those two apart and you can read any vendor case study in this industry.

Reading vendor claims without being taken in

The weak end of the same list. The three cases above came with named plants and published numbers; these do not.

Adani Power AVEVA PI programme	8 sites, 25 units, 150,000 tags, live screens — and no published benefit figure anywhere, including their own annual report	Proven as deployed; benefit unquantified publicly
NTPC NePPS operator advisory	Named system, named research centre, no published benefit figure	Proven as deployed; benefit unquantified publicly
GE Vernova performance software	'Up to 1 % heat rate reduction in 12 months'; customer examples anonymous	Emerging — vendor claim, no named plant
Tata Power enterprise AI platform	Announced April 2026; grid and renewables, thermal not named	Emerging — a platform announcement, not a result
Generic 'AI saves 10–15 %' marketing	No plant, no baseline, no measurement method	Experimental — treat as advertising until a baseline appears

Three questions for any vendor: which plant, what baseline, and who measured it? If any answer is missing, the number is marketing.

What the evidence actually supports

Setting an expectation you can defend in a review meeting

WHAT THE BROCHURES SAY

5 to 15 per cent

Efficiency gains, availability improvements and cost reductions quoted without a named plant, a stated baseline or an independent measurement. Almost always a pilot result on a poorly tuned unit, annualised and generalised.

WHAT THE PUBLISHED FLEET RESULTS SAY

About 1 per cent

Vistra's 67-unit rollout averaged 1 per cent against a 2 per cent pilot. GE Vernova claims up to 1 per cent in twelve months. On MAHAGENCO's June 2026 thermal generation, 1 per cent of heat rate is roughly 24 kcal/kWh on a 2,400 kcal/kWh unit — real money, and nothing like the brochure.

Plan on the fleet number, not the pilot number. A business case built on 10 per cent will be shredded, and it will take the credible 1 per cent down with it.

SESSION

Operation and performance optimisation

Combustion, condenser, auxiliary power and coal quality — where the kcal/kWh actually come from

- The controllable-loss table, and online loss accounting
- Combustion optimisation and the engineering that makes it safe
- Auxiliary power as a scheduling problem
- How a soft sensor is built

Hands-on: SIM-6 and SIM-3

15:12 – 15:45

33 minutes

The controllable loss table

Indicative heat-rate sensitivities — subcritical against supercritical, and who can actually move each one

Parameter	Deviation from design	Subcritical 210–500 MW	Supercritical 660 MW	Who influences it daily
Boiler efficiency	1 % drop	≈ 30 kcal/kWh	≈ 25 kcal/kWh	Operations and combustion
Main steam temperature	10 °C below design	5–8 kcal/kWh	4–6 kcal/kWh	Operations
Reheat steam temperature	10 °C below design	4–6 kcal/kWh	4–5 kcal/kWh	Operations
Condenser back pressure	10 mmHg rise	8–12 kcal/kWh	7–10 kcal/kWh	Mechanical and chemistry
Excess O ₂	1 % above optimum	10–15 kcal/kWh	8–12 kcal/kWh	Operations, every shift
Unburnt carbon in ash	1 % increase	10–15 kcal/kWh	10–15 kcal/kWh	Mills and combustion
Final feedwater temperature	5 °C below design	4–6 kcal/kWh	4–6 kcal/kWh	Heaters and mechanical
APH air leakage	5 % increase	3–5 kcal/kWh, plus ID fan power	3–5 kcal/kWh, plus ID fan power	Maintenance

**₹14–44
lakh a year**

what one kcal/kWh is worth on a
210 to 660 MW unit — indicative,
at 65 per cent PLF

Online loss accounting

What an expected-heat-rate model does that a monthly spreadsheet cannot

THE MONTHLY PERFORMANCE REPORT

One number, four weeks late

It tells you the unit ran at 2,425 kcal/kWh against 2,375 normative. It cannot tell you which shift, which coal, which mill combination or which fouled surface produced the gap — because by the time it is compiled, every one of those conditions has gone.

AN EXPECTED-HEAT-RATE MODEL

A deviation, attributed, now

It computes every few minutes the heat rate this unit should be achieving at this load, this coal, this ambient and this cooling water temperature, then splits the gap across back pressure, excess air, unburnt carbon, feedwater temperature and auxiliary power while the shift is still running.

A monthly report tells you that you lost the money. An online loss account tells you where it went, and who is still in a position to stop it.

Combustion optimisation — what it actually does

And what has to be engineered around it before it goes anywhere near the DCS

What the system does

- Learns a boiler model from your own operating data, not from a design manual
- Searches combinations of excess O₂, burner tilt and secondary air damper positions
- Brings mill combination and over-fire air into the same search
- Writes small bias values to the DCS — it nudges setpoints, it does not replace the control loops
- Re-optimises as coal, load and equipment condition change through the shift
- States the predicted gain in kcal/kWh before anything is allowed to move

What honest engineering requires

- Hard clamps on every variable it is permitted to touch
- Rate limits, so nothing moves faster than the boiler can follow
- A watchdog that returns the unit to base control if the optimiser stops responding
- Bumpless transfer in and out, tested in both directions
- Operator override on one button, available at all times, no password
- Availability above 90 per cent in service before anyone believes the reported benefit

Vendors sell the left column. Your commissioning engineer has to build the right one — and the right one decides whether the benefit still exists in year two.

The trade-off nobody puts on the brochure

NOx, unburnt carbon, heat rate and CO move against each other — always

● NOx

Lower excess air and staged combustion bring NOx down. The same moves push unburnt carbon up and can cost flame stability at low load.

● Unburnt carbon

Each 1 per cent increase costs an indicative 10 to 15 kcal/kWh and downgrades the ash. Raise O₂ to fix it and you pay in NOx and dry gas loss.

● Heat rate

The best O₂ for heat rate sits above the best O₂ for NOx and below the point where CO breaks away. It is a narrow window and it moves with coal.

● CO and slagging

Rising CO is the boiler saying the window has closed. Slagging and high furnace exit gas temperature are the slower, more expensive version of the same message.

The weights in that objective function are a management decision, not a technical one. Somebody senior has to sign what a milligram of NOx is worth in kcal/kWh.

Lab — SIM-6 · Combustion multi-objective optimiser

Twelve minutes at the keyboard, on the trade-off surface you have just been shown

SIM-6

Combustion multi-objective optimiser

12 minutes at your laptop

Do this

1. Open SIM-6 and run the optimiser at its default weights. Write down the proposed excess O_2 , the burner tilt and the predicted gain in kcal/kWh.
2. Drag the weight on NOx to maximum and run again. Watch the recommended target slide along the frontier and the heat-rate gain shrink.
3. Bring the NOx weight back down and settle on a setting you would be willing to defend to your environment cell.
4. Switch the hard clamps off and re-run. Read what the unconstrained optimiser proposes, and decide whether you would sanction it on a running unit.
5. Switch the rate limits off and watch the target jump in one step instead of walking there.

Watch for

- There is no single best answer — only a frontier, and a choice about where on it you wish to sit.
- Weighting NOx heavily is a perfectly legitimate answer. It costs heat rate, and the cost is on the screen in kcal/kWh.
- With clamps off, the optimiser asks for an operating point no engineer in this room would sign for.
- With rate limits off, the setpoint steps instead of ramping — which is precisely how a boiler gets upset.

Combustion optimisation is a trade-off surface, not a single best answer. Clamps and rate limits are not paperwork — they are the engineering that makes an optimiser safe enough to leave in service.

Condenser, vacuum and cooling

The cheapest kcal on the station, and the one most often lost quietly

Telling fouling from air ingress

- Tube fouling: terminal temperature difference rises while the cooling water temperature rise falls
- Air ingress: terminal temperature difference rises with the water side holding, and dissolved oxygen follows
- Cleanliness factor computed continuously, not once a year on the day of the performance test
- Vacuum pump or ejector duty trended against estimated condenser air load
- Ball cleaning judged by the cleanliness factor it delivers, not by the ball counter
- Back pressure always read against what is achievable at today's wet bulb

What you can actually change this week

- CW pump combination against cooling water inlet temperature and unit load
- Cooling tower performance measured as approach to wet bulb, not as outlet temperature
- Fill condition, drift and water distribution — the slow losses nobody logs
- Tube cleaning scheduled by cleanliness factor rather than by calendar
- Air ingress hunted before the planned outage, so the leak is known when the unit opens
- Condenser duty checked at part load, where most of the month is actually spent

A 10 mmHg improvement in back pressure is worth an indicative 8 to 12 kcal/kWh on a subcritical unit. Nothing else on the station is that cheap.

Auxiliary power is a scheduling problem

81.5 MU above norm in June 2026, worth ₹32.9 crore — all of it consumed inside the fence

● Mill combination

Which mills run, at what loading, on which elevations. Habit picks the combination. A search picks a better one, and re-picks it when the load moves.

● Fan control mode

Inlet guide vanes against variable speed, and how hard the ID fan is throttling against a damper somebody set years ago and nobody has revisited.

● CW pump scheduling

Pump count against cooling water inlet temperature and load. The right number in February is the wrong number in May, and the schedule rarely changes.

● ESP field energisation

Fields run at reduced power or de-energised when ash loading genuinely permits, inside the emission limit and continuously verified against opacity.

● Compressed air

Leak load, unloaded running hours and receiver pressure. The cheapest auxiliary saving on any station, and by some distance the least glamorous.

● Ash handling cycles

Cycle frequency matched to actual ash generation rather than to a fixed timer that was set at commissioning and never questioned since.

Chandrapur 3-7 at 11.69 % against a norm of 8.67 %. Parli 6-7 at 12.41 % against 9.30 %. Nashik at 12.96 % against 10.75 %. None of that is a plant defect.

Lab — SIM-3 · Auxiliary power optimiser

Ten minutes at the keyboard, on your own station's typical operating point

SIM-3

Auxiliary power optimiser

10 minutes at your laptop

Do this

1. Open SIM-3. Set the unit load and the cooling water inlet temperature to the values your own station typically runs at this time of year.
2. Set the mill, fan, CW pump and ESP configuration you believe your unit is running right now.
3. Read the configuration the optimiser proposes against the one you entered, and check that every constraint is still satisfied.
4. Press 'Adopt the optimiser's answer' and read the saving, in MU and in rupees a month.
5. Raise the cooling water temperature to 38 °C, re-run, and watch the optimum CW pump count change.
6. Give yourself two minutes to beat the optimiser by hand.

Watch for

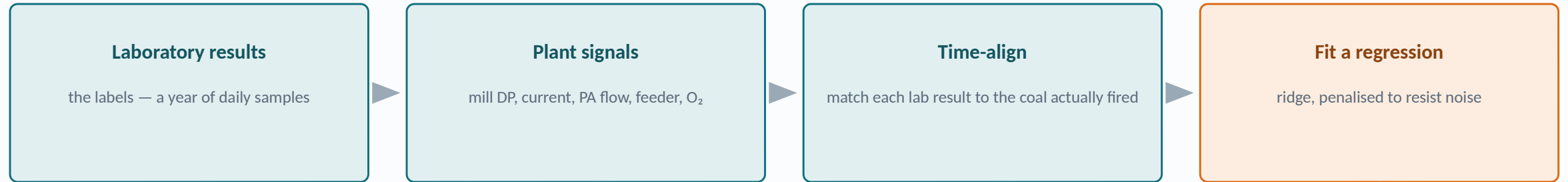
- The habitual configuration is rarely the cheapest one, and it is never the cheapest one at every load.
- The optimum CW pump count moves with cooling water temperature. A February schedule is simply wrong in May.
- The saving appears in MU and in rupees at your station's own variable charge, not as a percentage.
- Beating a combinatorial search by hand is possible occasionally and impossible consistently.

Auxiliary power is a combinatorial scheduling problem, and a computer solves it better than habit does. The 81.5 MU the fleet burned above norm in June 2026 is largely made of exactly these choices.

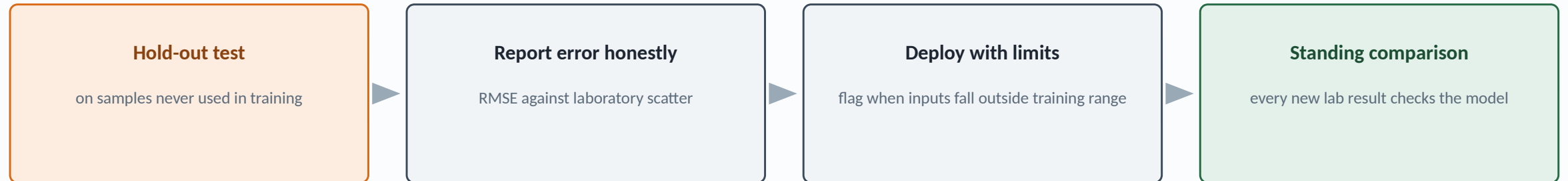
How a soft sensor is built

Using the coal GCV estimator as the worked example — the same shape applies to any inferred measurement

BUILD



OPERATE

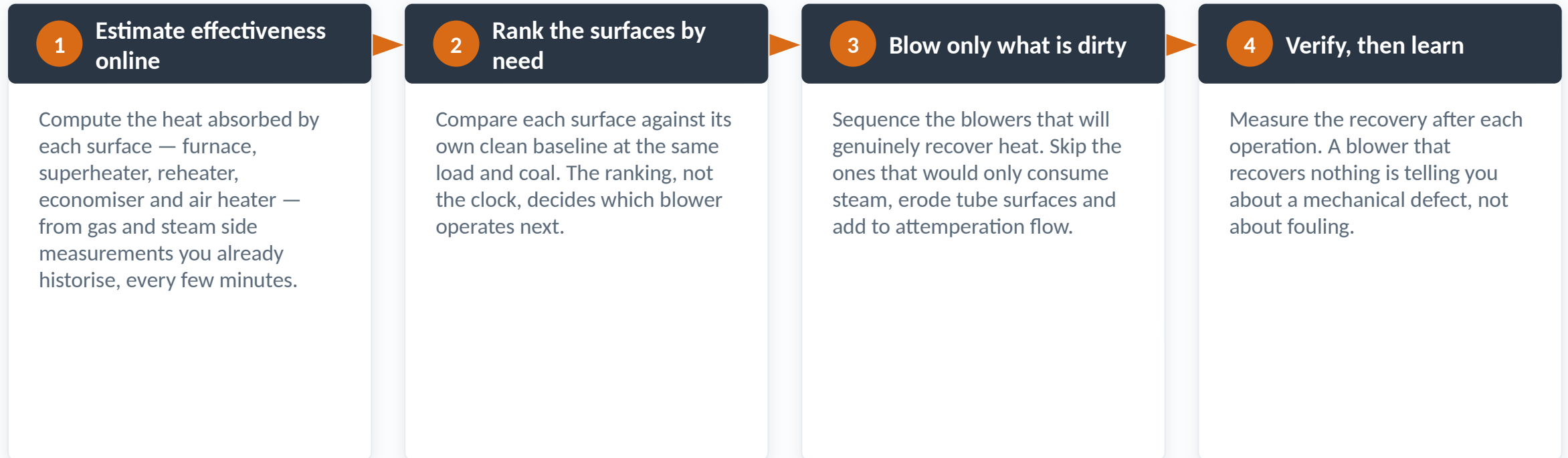


Feedback: Retrain on a change of coal source — the single biggest failure mode of a GCV soft sensor

A soft sensor never beats the laboratory it learned from. It interpolates between laboratory results — which is exactly what you need between them.

Soot blowing on demand, not on a timer

The heat transfer surfaces will tell you when they need cleaning — if anyone is listening



Timer-based blowing wastes steam on clean surfaces and leaves the dirty ones alone. Both cost heat rate, and one of them also costs tube life.

Coal quality and the GCV soft sensor

As-fired GCV ranges from 2,733 to 3,270 kcal/kg across our own fleet

● The gap the laboratory leaves

Bomb calorimeter results arrive hours or days later, on one sample, per rake. The mills are burning something entirely different by the time the figure lands.

● What a soft sensor does

Estimates as-fired GCV every few minutes from mill power, air flow, feeder rate, O₂ and steam generation — trained on the laboratory results themselves.

● Blend against constraints

Chooses the blend against mill capacity, ESP loading and emission limits, not against price alone. A cheaper coal that de-rates a mill is not cheaper.

● Transit-loss analytics

GCV loss between loading and unloading runs 619 to 941 kcal/kg. Rake-level analysis separates sampling error from genuine loss, rake by rake.

Coal is the largest line in the bill, and it is also the one number in the plant that almost nobody knows while it is being burnt.

What an advisory looks like at 14:20

A 500 MW unit, the deviation attributed, and a rupee figure attached to the recommendation

Chandrapur Unit 8 · 500 MW · Performance advisory · 14:20

● Net heat rate	2,425 kcal/kWh against 2,375 normative — plus 50
● Contributor 1 · boiler and turbine	Gross heat rate 2,258 against 2,232 — worth about 27 kcal/kWh
● Contributor 2 · auxiliary power	6.90 % against a 6.00 % norm — worth about 23 kcal/kWh
● Largest single item	Condenser back pressure above its wet-bulb-corrected expectation — about 11 kcal/kWh
● Recommended action	Run condenser ball cleaning this shift; review CW pump count at present cooling water temperature
● Value if actioned	About 8 kcal/kWh — roughly ₹27 lakh a month on this unit at its own cost of heat

Advisory means the operator decides. Closed-loop means the system acts within limits you set. Know which one you are buying.

Where the money is at your station

Block five close — and the answer is different at every one of the thirteen groups

● If auxiliary power is your whole problem

Nashik, Khaperkheda 5, Chandrapur 3-7 and Parli 8 all beat their gross heat-rate norm. Every kcal of their penalty is auxiliary. Start with SIM-3.

● If it is boiler and turbine

Koradi 8-10 carries a 176 kcal/kWh gross gap, Paras 3-4 89 kcal/kWh and Koradi 6 56 kcal/kWh. Start with combustion, condenser and air heater.

● If it is both

Bhusawal, Khaperkheda 1-4, Chandrapur 8-9 and Parli 6-7 carry both. Do the decomposition first and stop guessing which half is the larger one.

● The Monday version

One unit, one month of historian data, one decomposed loss account. No purchase order and no vendor is required to produce that.

₹56 crore of net heat-rate gap and ₹33 crore of excess auxiliary power in a single month. Both are addressable with data this company already owns.

SESSION

Generative AI, RAG and agents

Turning thirty years of manuals, SOPs and RCA reports into something you can ask a question

- What a language model is, and why it is confident when wrong
- The RAG pipeline, end to end, with citations
- From assistant to agent — the reason-act-observe loop
- Where it will let you down, and what to never use it for

Hands-on: SIM-7 and SIM-8

15:55 – 16:28

33 minutes

What a language model actually is

Next-token prediction — and why it sounds most confident exactly when it is wrong

WHAT IT IS — A MODEL OF LANGUAGE

A very good guess at the next word

It was trained on an enormous quantity of text to predict what word comes next. Everything it does — summarising, drafting, explaining, translating between Marathi and English — is that single operation repeated. It has learned the shape of technical writing extremely well.

WHAT IT IS NOT — A SOURCE OF TRUTH

It has no idea what is true

There is no database inside it and no lookup step. A plausible sentence and a correct sentence feel identical to it, so it produces both in the same fluent, confident, well-formatted English. Fluency is not evidence. Confidence is not accuracy.

A brilliant assistant and a terrible authority. Treat every unsourced answer as a draft written by somebody who has never seen your plant.

The plant knowledge problem

Everything you need to answer a question at two in the morning, and none of it findable

● OEM manuals

Thousands of pages per package, across several vendors and several decades. Searchable only by whoever remembers which volume it was in.

● SOPs and operating instructions

Written, revised, superseded. The current version and three obsolete ones sit in the same folder with almost the same file name.

● RCA and trip reports

The most valuable documents on the station. Written once, filed, and almost never read again by anyone outside the original committee.

● Defect and work-order history

Years of CMMS free text. Every fault this station has ever had is in there, in a form nobody can count or sort.

● Drawings and protection settings

P&IDs, single-line diagrams, relay settings, interlock schemes — the highest-consequence documents, held under the strictest version control.

● Tacit knowledge

The engineer who knows why that damper has been at 40 per cent since 2014. That knowledge walks out of the gate on the day he retires.

None of this is a data problem. All of it is a language problem — and language is the one thing this technology is genuinely good at.

Three ways to make a model know your plant

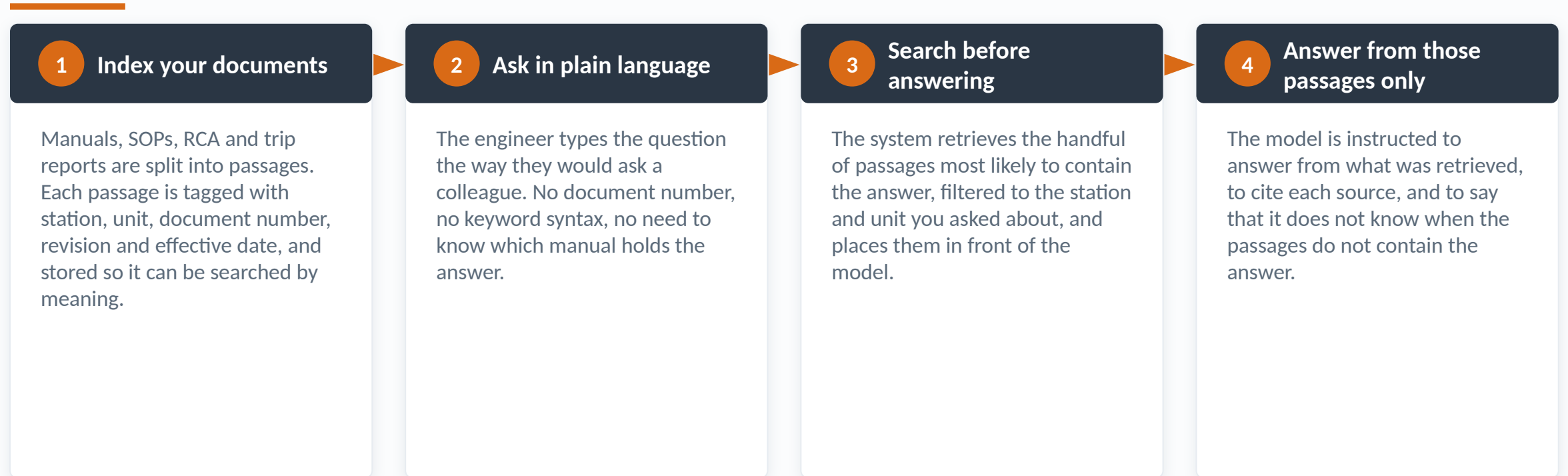
Prompting, retrieval and fine-tuning — and only one of them belongs in a power station

Approach	What it needs	When knowledge updates	When it is the right choice
Prompting	You paste the relevant text into the question yourself	Only what you pasted, only this time	One-off questions on a document already open in front of you
Retrieval (RAG)	Your documents, indexed and tagged with unit, revision and date	The moment you re-index a revised document	Almost always — this is the answer for a power station
Fine-tuning	Thousands of curated question-and-answer pairs, and GPU time	Never, until you retrain and re-validate the whole model	Almost never — for tone and format, not for facts

Fine-tuning teaches a model how to speak. Retrieval teaches it what is true today. Your revision-control problem is a retrieval problem.

RAG, explained properly

Search first, then answer — and show the passage the answer came from



An answer you cannot trace is an answer you cannot use. The citation is not decoration — it is the entire control.

Metadata matters more than the model

The failure is never 'the AI was stupid'. It is 'the AI answered about the wrong unit'.

● Station and unit on every chunk

Koradi Unit 8 and Koradi Unit 10 are not the same machine. Without a unit tag the retriever cannot tell them apart, and it will not try.

● Revision and effective date

A superseded SOP retrieves exactly as well as the current one. Unless the revision is on the passage, the model quotes whichever reads better.

● Document class and status

Manual, SOP, trip report, drawing, circular — and current, superseded or withdrawn. A withdrawn procedure must never surface as an answer.

● The concrete failure

You ask about the Unit 4 boiler feed pump and receive the Unit 3 answer: correctly cited, fluently written, and completely wrong for your machine.

Ninety per cent of the engineering in a plant retrieval system is metadata and version control. The model is the easy part.

Lab — SIM-7 · Retrieval over plant documents

Ten minutes at the keyboard, on an indexed corpus of plant documents

SIM-7

RAG over plant documents

10 minutes at your laptop

Do this

1. Open SIM-7 and run the default ID fan question. Before you read the answer, read the retrieved passages and their similarity scores.
2. Now read the answer and check every sentence of it against the passages it cites.
3. Switch 'Grounded' off and ask the same question. Compare the invented answer with the cited one, line by line.
4. Turn hybrid search off and search for a document code. Watch pure meaning-based search fail on an exact string.
5. Filter by station and unit, then re-run a question that was previously answered from another unit's document.

Watch for

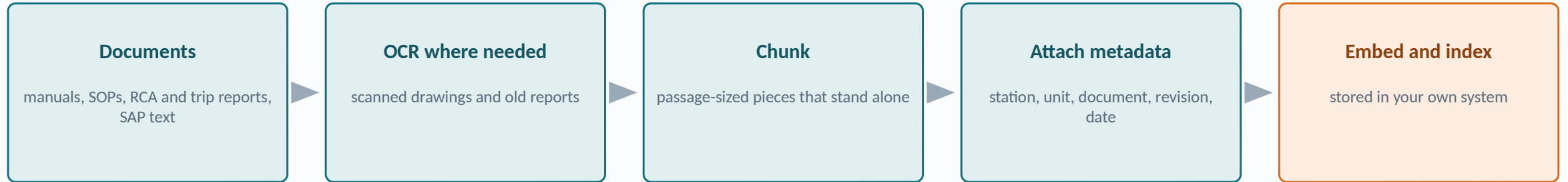
- The retrieval step happens first and is completely visible. Nothing is magic once you can read the passages yourself.
- Ungrounded, the model writes a confident, plausible, well-structured answer with no source and at least one invention in it.
- Meaning-based search alone cannot find an exact document code. Keyword matching is still needed alongside it.
- The unit filter is the thing that stops a Unit 3 passage answering a Unit 4 question.

The difference between an ungrounded chatbot answer and a retrieval-grounded answer with citations is not a matter of degree. One can be checked in thirty seconds. The other cannot be checked at all.

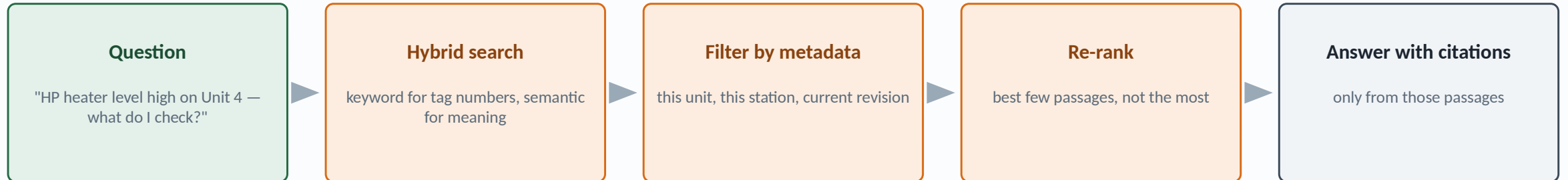
The RAG pipeline, end to end

What happens between an engineer's question and a cited answer

INDEXING — once per document, then on revision



ANSWERING — every question



Feedback: A superseded revision must leave the index the day it is superseded — version control is your job, not the model's

The language model appears in exactly one box, at the very end. Everything that determines whether the answer is trustworthy happens before it.

Practical uses in our own workflow

Eight things worth trying, none of which touch a control system

● Shift handover drafting

Drafted from the log, the alarm list and work in progress. The outgoing engineer corrects it and signs it. Ten minutes recovered, every shift.

● Trip report first drafts

Sequence of events, protections operated and plant response assembled into a first draft within the hour, while the committee is still being formed.

● Troubleshooting assistant

Describe the symptom in plain language and receive the relevant manual sections, past defects and previous RCAs, with citations, in one place.

● Precedent search across the fleet

Has this failure happened at another station? Today that question needs a phone call and some luck. It should need a search box.

● Work-order text into a taxonomy

Years of free-text CMMS entries classified by equipment, failure mode and cause — turning unreadable text into countable reliability data.

● Specification drafting

First-cut technical specifications and tender clauses assembled from your own earlier documents, so house style and referenced standards survive.

● Marathi and English safety communication

Toolbox talks, safety circulars and site instructions in both languages, so nothing important is understood only by those who read English.

● Summarising inspection reports

Overhaul and inspection reports condensed into findings, actions and open points, with every point traceable to the paragraph it came from.

Every one of these produces a draft for an engineer to check. None of them produces a decision. That distinction is the whole safety case.

Why we cannot explain 57.85 per cent

Tying language back to the June 2026 numbers

THE RELIABILITY QUESTION

Why was availability 57.85 %?

Khaperkheda Units 1-4 ran at 57.85 per cent availability in June 2026 and Chandrapur 3-7 at 64.88 per cent. Between them they carry ₹45.69 crore of the fixed-charge disallowance. Nobody can currently say which failure modes account for how much of it.

WHY WE CANNOT ANSWER IT

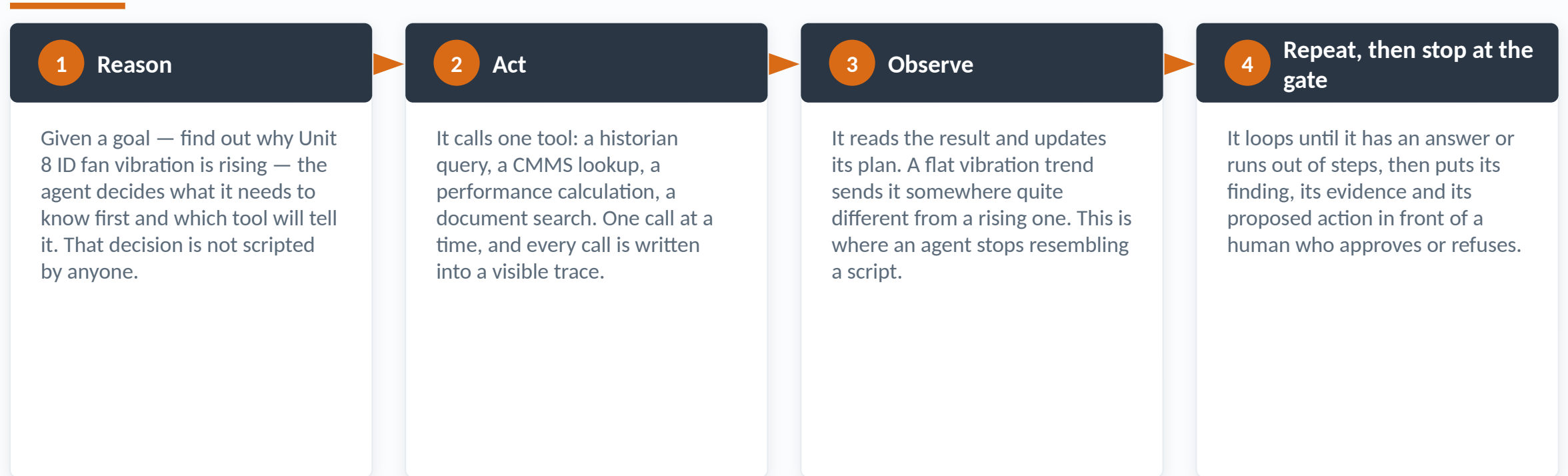
The history is free text

Every one of those outages is recorded — in work orders, log books and trip reports, written by different people in different words over many years. You cannot count what has never been classified, and a Pareto chart of unclassified text does not exist.

Structuring that history is a language problem, not a reliability problem. Solve the language problem and the reliability analysis becomes ordinary engineering.

From assistant to agent

A model given tools, a goal, and permission to decide the order of the calls



A chatbot answers. A script follows a fixed path. An agent chooses the path — which is precisely why it must stop at a human gate before anything is actioned.

What a tool is, for a power station

An agent is only as safe as the narrowest tool you hand it

Tool	What it does	Access level	Why it is drawn this way
Historian query	Fetches tag trends for a plant item over a stated period	Read-only	No mechanism exists to write anything back to the historian
CMMS lookup	Finds defect history, open work orders and spares for that item	Read-only	Raising or closing a work order stays a human act
Performance calculation	Runs your own heat-rate and loss model on the retrieved data	Compute only	Touches no plant system and holds no plant credentials
Document search	Retrieves manual, SOP and RCA passages with citations	Read-only	The corpus is a copy; the agent cannot alter a document

Read-only is safe. Write-capable — a report generator that files the report, a tool that raises the work order or moves a setpoint — is a permission, not a convenience, and it needs a human at the gate every single time.

Lab — SIM-8 · Agentic work-order assistant

Ten minutes at the keyboard, on the Koradi Unit 8 ID fan scenario

SIM-8

Agentic AI work-order assistant

10 minutes at your laptop

Do this

1. Open SIM-8 and start the Koradi Unit 8 ID fan scenario. Step it forward one tool call at a time — do not press run-all.
2. At each step read the agent's stated reasoning before you look at what the tool returned, and decide whether you would have made the same call.
3. Continue to the approval gate. Read the proposed work order, its evidence and its priority, then approve or refuse it yourself.
4. Tick 'grant write access' and read the warning that appears. Do not skim it.
5. Re-run with one tool disabled and watch the agent's plan change around the missing information.

Watch for

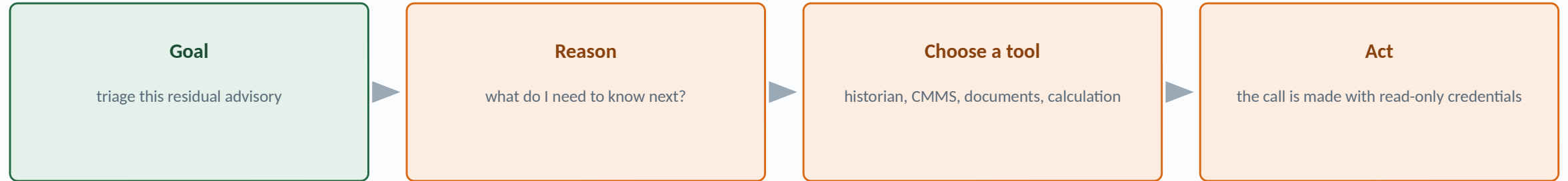
- The plan is not fixed. What the historian returns changes which tool the agent reaches for next.
- Every call, every result and every inference is on the screen. An agent whose trace you cannot audit is not deployable.
- The approval gate is where engineering judgement lives, and it is the only place it lives.
- With write access granted, the same trace that was advice becomes an instruction. The warning is not boilerplate.

An agent is a planner that calls tools. The visible trace is what makes it reviewable and the human approval gate is what makes it deployable. Neither of the two is optional.

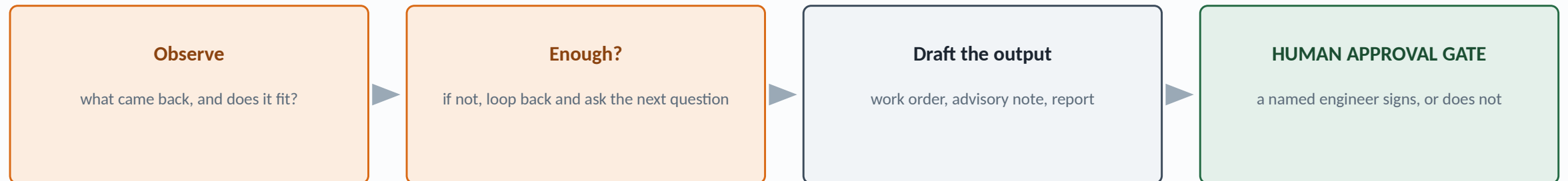
The agent loop

Reason, act, observe, repeat — and where the loop is required to stop

THE LOOP — repeats until the agent has enough



THE STOP



Feedback: Every loop iteration and every tool call is logged, so the reasoning can be reconstructed afterwards

The loop is what makes it an agent. The gate is what makes it acceptable in a power station.

Where it will let you down

Five failure modes, every one of them seen in practice, none of them fixed by a better model

Confident and wrong

It will invent a clause number, a setting or a torque value in fluent English and cite nothing at all. Grounding and citations reduce this sharply. Nothing eliminates it.

Version and unit drift

A superseded SOP or another unit's manual retrieves just as well as the correct one. Without revision and unit metadata you get correct answers to the wrong question.

It cannot do sums

A language model predicts text, not arithmetic. Any heat rate, efficiency or rupee figure must be produced by a calculation tool and checked — never lifted from the prose.

Prompt injection

Text inside an ingested document can carry instructions the model will follow. A scanned circular is not a trusted input, and an agent with write access makes that dangerous.

Confidentiality

Plant drawings, protection settings, tender papers and commercial data leaving the company through a public service is a breach, whatever the privacy statement claims.

Never use a language model for: protection settings • clearance and permit decisions • safety-critical procedure changes • statutory returns without verification • anything reaching the plant without an engineer's signature

What to try on your own documents

Block six close — four things that need no purchase order

● Pick one corpus, not all of them

One package's manuals, its SOPs and its last five RCAs. Small, current and version-controlled. A corpus you can trust beats a large one you cannot.

● Fix the metadata before the model

Station, unit, document number, revision, effective date, status. Tag fifty documents by hand and you will learn more than from any vendor demonstration.

● Test with questions you already know

Ask twenty questions whose answers you know, from your own documents. Count how many come back cited and correct. That number is your business case.

● Structure one year of work orders

Take one unit's CMMS free text and classify it by failure mode. The Pareto chart at the end is the one nobody in this company currently has.

Retrieval over your own documents is the cheapest, safest and fastest-returning AI project available to this company — and it begins with a filing exercise, not a model.

8

SESSION

Computer vision and emerging applications

Thermal, CCTV, drones and acoustics — and an honest maturity rating for everything else

- Vision in the plant, from switchgear to conveyor belts
- What a vision model needs from you
- Acoustic and ultrasonic methods
- Proven, Emerging, Experimental — and how to tell

16:28 – 16:40

12 minutes

Computer vision in the plant

Six places a camera earns its keep — and one boundary that decides whether the programme survives

● Thermal on electrical plant

Switchgear, busduct joints, cable terminations and transformer bushings. A loose or degrading connection appears as heat long before it fails.

● Thermal on boiler and coal

Casing and duct insulation loss, refractory condition, air preheater hot spots, and spontaneous combustion developing inside the coal stockpile.

● Conveyors and CHP

Belt tear and misalignment, chute spillage, idler condition and foreign objects — the failures that stop coal reaching the bunkers.

● Leak detection

Steam and oil leaks picked up on camera, including areas people rarely walk: roof levels, bunker floors and duct runs behind plant.

● Drones and crawlers

Chimney and flue liner, cooling tower shell, ash dyke survey, and stockpile volume by photogrammetry. Subject to DGCA permissions.

● Safety and zones

Restricted-area intrusion, hot-work monitoring and man-down detection. Monitor zones and equipment — never individual behaviour.

Monitor zones and equipment, not individuals. Behaviour monitoring raises privacy and industrial-relations issues that will sink the programme before it delivers anything.

What a vision model actually needs from you

The unglamorous work that decides whether a camera project delivers or embarrasses you

What the model needs

- Labelled images — several hundred to a few thousand, marked by somebody who knows the plant
- Examples of the fault, not only of normal — and faults are rare by definition
- Camera placement fixed, framed on the asset, and not moved after training
- Lighting that is consistent, or a thermal camera that does not care about light
- Lens cleaning, condensation and coal dust treated as a maintenance task with an owner
- A named engineer to review and correct wrong detections weekly for the first few months

What people get wrong

- Assuming a model bought for another station will work here — it will not
- Different camera, angle, lighting or equipment means re-labelling, not reconfiguration
- Expecting the demonstration accuracy on your own dusty, back-lit, monsoon images
- Installing cameras first and deciding what to detect afterwards
- No plan for who acts on a detection at two in the morning
- Treating labelling as a one-off, when the plant and the model both keep changing

A model trained at one station will not transfer to another without re-labelling. Transfer learning shortens that work; it does not remove it.

Acoustic and ultrasonic

Listening to the plant — the oldest diagnostic sense, now instrumented and trended

● Boiler tube leak detection

Acoustic emission sensors on the waterwall and second pass hear a leak as rising high-frequency energy, often days before the unit trips.

● Cross-check with a mass balance

Pair the acoustic signal with a data-driven makeup water mass balance. Two independent indications turn a suspicion into a defensible outage decision.

● Steam and air leak surveys

Ultrasonic detectors find compressed air and steam leaks that a walkdown misses. Auxiliary power and heat-rate loss, both directly costed.

● Acoustic pyrometry

Time-of-flight of sound across the furnace builds a temperature map where no thermocouple survives. Useful for combustion and burner balancing.

The acoustic signal tells you something is leaking. The mass balance tells you how much. Neither on its own justifies pulling a unit off bars.

Emerging applications, honestly rated

A filter to apply to every vendor presentation you attend for the next two years

Acoustic boiler tube leak detection	Acoustic emission sensors alarmed on rate of change, confirmed against a makeup water mass balance	Proven — with a mass-balance cross-check
Thermal and CCTV analytics	Fixed cameras and detection models on electrical plant, conveyors, leaks and restricted zones	Proven for thermal • Emerging for video
Drone and crawler inspection	Aerial and crawling platforms carrying camera and thermal payloads where people cannot safely go	Proven — subject to DGCA permissions
Acoustic pyrometry	Time-of-flight of sound across the furnace reconstructs a temperature map for combustion balancing	Emerging — few installations in India
Digital twin	A calibrated hybrid of physics and data, kept live and honest against the running unit	Emerging — heavy engineering to sustain
Agentic assistants	A planner that calls historian, CMMS and performance tools, with a human approval gate	Experimental — approval gate not optional
Reinforcement learning for unit control	An agent learning a control policy by trial, which means learning from failures you cannot allow	Experimental — not on a live unit

Buy Proven. Pilot Emerging. Read about Experimental — and be suspicious of anyone selling it to you today.

Block close — vision and emerging

Four things to carry into the last session of the day

● Cameras are cheap; labels are not

The cost of a vision project sits in labelling, mounting, lighting and weekly correction of wrong detections. It does not sit in the model.

● Nothing transfers for free

A model trained at one station needs re-labelling at the next. Budget for it, or the second site will fail and the programme will be cancelled.

● Zones and equipment, never people

Individual behaviour monitoring will end the programme. Keep the scope on plant condition and on restricted areas, and say so in writing.

● Two defensible starts

Fixed thermal on electrical plant, and acoustic tube-leak detection cross-checked against a makeup water mass balance. Both produce a number an engineer can check.

Buy Proven. Pilot Emerging. Read about Experimental — and ask any vendor which one they are selling.

SESSION

Implementing AI responsibly

Data quality, the safety boundary, governance — and choosing one pilot you can actually finish

- Three rules: data quality, the safety and cyber boundary, human in the loop
- Governance for a state utility
- Why programmes fail, and sizing the prize honestly
- One pilot per station, from your own June 2026 evidence

16:40 – 17:00

20 minutes

Rule one — data quality and engineering validation

Before the algorithm, the tag list. This is the least glamorous slide of the day and the most important.

What will bite you

- Historian compression and exception deviation quietly discarding the excursion you needed
- Frozen transmitters reading a flat, plausible value for months on end
- Calibration drift learned by the model as normal behaviour
- An asset hierarchy that does not match between DCS, historian and SAP
- Tags that exist in the DCS but were never historised at all
- Timestamp misalignment between systems, so cause appears to arrive after effect

What good practice looks like

- Audit the tag list before the first model, not after the first failure
- Check compression settings on every tag the model will actually use
- Validate every model output against first-principles engineering
- Reconcile the asset hierarchy once, properly, across DCS, historian and SAP
- Keep a named owner for the tag list — a person, not a department
- Treat data quality as permanent maintenance, not a project phase

Most plant AI projects fail on the asset hierarchy, not on the algorithm.

Rule two — the safety and cyber boundary

Where AI may go, where it may go only with engineering controls, and where it must never go

NEVER

Protection systems, trip logic, interlocks and any safety instrumented function. No model, no advisory, no exception, at any station.

WITH CONTROLS

Closed-loop optimisation writing biases to the DCS — with clamps, rate limits, a watchdog, bumpless transfer, operator override and documented approval.

FREELY

Advisory systems, anomaly alerts, performance advisories, decision support and knowledge assistants. Validate the output; the engineer still decides.

Segmentation

Purdue levels with IEC 62443 zones and conduits, drawn and agreed on paper. The historian sits in the demilitarised zone. Nothing on the plant network talks directly to a corporate or cloud analytics service.

Read-only by default

Every AI system starts with read access to process data and nothing else at all. Write access back to the DCS is a separate, engineered, approved decision — never a setting somebody quietly changes.

One-way data paths

Where data must cross from OT to IT, use a unidirectional gateway or a data diode. Physics enforces the direction of flow, not a firewall rule that can be edited.

OEM remote access

The single largest OT risk at most stations, and usually nobody's job to manage. It must be time-boxed to the visit, logged, supervised by a station engineer, and disabled by default between sessions.

AI-specific risks

Four risks that are genuinely new: plant data leaving the fence, model poisoning through corrupted history, prompt injection hidden inside a document the assistant reads, and supply-chain risk in machine-learning packages.

Rule three — human in the loop

The system advises, the engineer decides. Both halves of that sentence have to stay true in practice.

● Trust calibration

Two failures. Under-trust, where nobody opens the alerts and the system quietly dies. Over-trust, where the operator stops thinking. The second is far more dangerous.

● Alert fatigue

Twenty alerts in the first fortnight and the system is finished. Start with three machines, tune persistence, and earn the right to add more.

● Workflow fit

If the output does not arrive in the shift log, the round sheet or the work order, it does not exist. A separate screen is a screen nobody opens.

● Accountability

The decision stays with the named engineer. Record what the system said, what the engineer decided, and why — especially when the two differ.

Under-trust is visible and recoverable. Over-trust is invisible until the day the model is wrong — and that is the day it matters.

Governance for a state utility

What has to exist on paper before a fleet programme — not after the first audit query

● An AI usage policy

One page, circulated. Permitted uses, prohibited uses, and who approves an exception. A forty-page policy is a policy nobody reads.

● Data classification

What is public, internal, confidential and regulated — and an explicit list of what may never be pasted into a public AI service.

● Verification requirement

Any AI-produced figure that leaves the station is checked by a named engineer against source data before it reaches a report or a filing.

● Record keeping and audit trail

Prompts, outputs, model versions and approvals retained. If a decision is questioned two years later, the trail has to exist.

● DPDP Act 2023

Where personal data is involved — CCTV, biometrics, employee records — consent, purpose limitation and retention are legal obligations, not preferences.

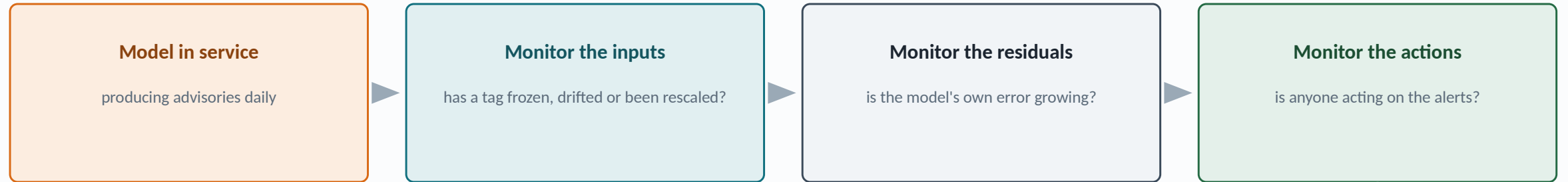
● Procurement questions

Where does our data go, who owns the model, what is the accuracy on our plant, what happens at contract end, and who sustains it.

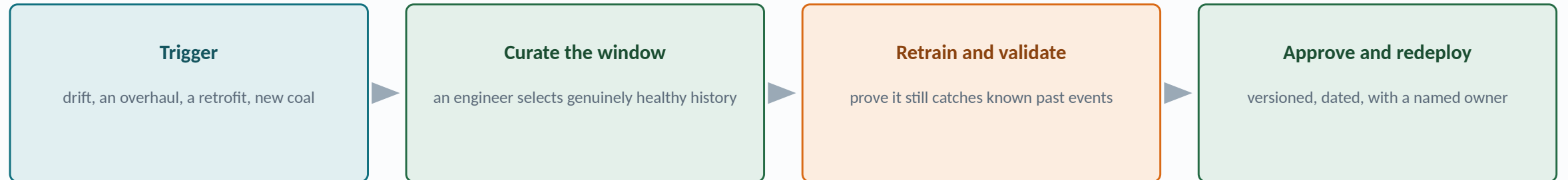
What happens after go-live

The lifecycle nobody budgets for, and the reason models quietly stop working

WATCHING



RENEWING



Feedback: Put 'retrain the models' in the outage checklist, next to 'recalibrate the instruments'

A model is a piece of plant. It degrades, it needs overhauling, and it needs an owner. Budget for it or lose it.

Why AI programmes fail

Eight failures seen again and again, and the countermeasure for each one

WHY IT FAILED

Started with technology, not a problem

Bought a platform, not an outcome

Pilot ran on and on with no decision

No baseline, so the benefit was disputed

Operators heard about it after go-live

Data promised by IT, never delivered

Scaled to the fleet before proving one unit

Nobody budgeted to run it after year one

WHAT PREVENTS IT

Write the problem and its rupee value before any vendor call

Contract on a validated result at one unit, not on a licence

Fix the go or no-go date and the decision-maker up front

Agree the baseline and how it is calculated on day one

Shift engineers in the design review from week one

Extract and inspect 12 months of history before signing

One machine, one owner, one proven result, then replicate

Fund support, retraining and tags in the same sanction

Every one of these is an organisational failure, not a technical one — which is good news, because you control all of them.

Sizing the prize honestly

The June 2026 evidence, converted into a first-year target that finance will accept

Evidence, June 2026	Value at stake	Where it comes from	Realisation	First-year target
Net heat-rate gap vs MERC norm	₹56.35 cr/month	13 thermal groups	15–25 %	₹8.5–14.1 cr/mth
— of which auxiliary power	₹32.91 cr/month	81.5 MU above norm	15–25 %	₹4.9–8.2 cr/mth
Availability disallowance	₹100.87 cr cumulative	₹32.93 cr in the June bill	15–25 %	₹15.1–25.2 cr
Heat rate, annualised	≈ ₹676 cr/year	If June is a typical month	15–25 %	₹101–169 cr/yr

₹101–169 cr

a defensible first-year heat-rate target at 15 to 25 per cent realisation of the June 2026 gap

Indicative basis

Values are the full gap against MERC normative performance for June 2026. No programme closes a gap entirely — apply a realisation assumption before putting any of this in a business case.

These are the values of closing the gap entirely. No programme does that, and a business case built on full closure will not survive the finance department.

Choosing your first pilot

Seven criteria, six candidates, and a ninety-day plan with a name against each phase

Data already historised

Problem clearly defined

Value at stake is material

Benefit is measurable

Named owner at the station

Result can be deployed

Low acceptance risk

Candidate pilot	Value	Risk	Verdict
BFP or ID fan anomaly detection	High	Low	Strongest first choice — data exists, no process risk
Mill health and choking prediction	High	Low	Close second — mills hurt availability and heat rate together
Auxiliary power optimisation	High	Low	Benefit visible on the meter, so nobody argues about it
Condenser performance and cleaning schedule	Medium	Low	Good second wave — needs sound vacuum instrumentation
Document assistant on SOPs and RCA reports	Medium	Low	No process risk at all — can run alongside the main pilot
Closed-loop combustion optimisation	High	High	Do this third or fourth. Never first.

A 90-day starter plan

Days 1–30

Audit the tag list. Pick one machine. Extract twelve months of history and inspect it properly. Name the owner before anything else starts.

Performance plus C&I Engineer

Days 31–60

Build and validate on history. Confirm the model would have caught known past events, and confirm it stays quiet through normal operation.

Performance Engineer with analytics support

Days 61–90

Run live in shadow mode. Triage every alert, record what was found, and report the benefit against the agreed baseline.

Maintenance Planner with station sponsor

One pilot per station, from your own evidence

Four stations, four different diagnoses, four different first projects — all from June 2026

Station	The evidence, June 2026	The pilot	Monday morning
Nashik 3-5	Aux 12.96 % vs 10.75 %; gross HR 18 kcal/kWh better than norm	Auxiliary power optimisation — the whole 50 kcal/kWh penalty is auxiliary	List the CW pump, mill and ESP field combinations actually run last month
Koradi 8-10	Gross HR 2,272 vs 2,096 — a 176 kcal/kWh gap on 964.80 MU	Boiler and turbine heat rate: combustion and condenser analytics	Pull a month of O ₂ , back pressure and unburnt carbon at unit level
Khaperkheda 1-4	Availability 57.85 %, the fleet's lowest; ₹21.75 cr disallowance	Predictive maintenance on the two machines that forced the outages	List every forced outage of the last 24 months by equipment
Chandrapur 3-7	Aux 11.69 % vs 8.67 %; secondary oil 3.23 ml/kWh vs 1.00	Auxiliary power, plus flame stability to avoid oil support	Get oil consumption split by unit and by start-up from the log

Four stations, four different problems, four different first pilots. There is no fleet-wide answer, and anyone selling one has not read your numbers.

What to do on Monday morning

Six actions that are free, take under an hour, and need nobody's approval

● Ask how many tags are historised

Call your C&I team with two questions: how many tags are historised, and at what compression and exception deviation setting.

● Pick one machine that hurt you twice

One machine that has forced an outage twice in two years. Not a system, not a fleet. One machine, with a name and a number.

● Check the history actually exists

Find out whether twelve months of clean history exists for that machine. If it does not, that is your first project.

● Ask who would own an alert

If an alert arrived at two in the morning tomorrow, who opens it, who inspects, who closes it? No name means no programme yet.

● Put your own numbers in the value table

Your station's PLF, heat rate, as-fired GCV and coal cost into the value table from the handout. The answer then belongs to you.

● Run SIM-2 on your own station

Open the simulation lab, select your station, and see for yourself whether your gap is auxiliary power or boiler and turbine.

None of these six needs a sanction, a vendor or a budget. They need one hour and one engineer who decides to ask.

PREDICT · DIAGNOSE · OPTIMISE · ASSIST

Four things AI does well in a power station — and one thing only you can do

What you can do on Monday morning

- 1 Ask your C&I team how many tags on your unit are actually historised, and at what compression setting
- 2 Pick one machine that has hurt you twice in the last two years
- 3 Find out whether twelve months of clean history exists for it
- 4 Ask who at your station would own an alert if one arrived tomorrow morning
- 5 Put your own PLF, heat rate, GCV and coal cost into the value table and see what 10 kcal/kWh is worth to you
- 6 Open the simulation lab and run SIM-2 on your own station's row

The mathematics is the easy part. The data, the ownership and the follow-through are the job.

Questions — including from Koradi, Khaperkheda, Bhusawal and Paras